

A Measurement-Sensitivity Audit Protocol for Administrative-Capacity Studies in CDBG-DR Disaster Recovery

Kai-Fa Lu
Jesse Andrews
Ali Nejat

Abstract

Administrative-capacity research in CDBG-DR commonly reports a single point estimate from a single specification. We propose a six-item measurement-sensitivity audit protocol whose deliverable is a specification-curve dashboard, not a pass/fail verdict. Demonstrating on 543 local administering jurisdictions (primary) from HUD DRGR/QPR records (2003–2023) linked to QCEW NAICS 925110 staffing proxies (80.6% BLS-suppressed) and CDC/ATSDR vulnerability indicators: the capacity-timeliness coefficient is not stably identified. The dashboard spans positive, near-zero, and negative estimates depending on suppression handling, sample scope, capacity operationalization, and analytical framework; sign reversals and bootstrap confidence intervals crossing zero confirm the instability. A companion Cox survival analysis is itself measurement-sensitive (null under time-varying specification, positive under baseline). Capacity findings from this measurement stack are not robust enough for benchmarking without the protocol's sensitivity checks.

1. A Measurement-Sensitivity Audit Protocol for Administrative-Capacity Studies in CDBG-DR Disaster Recovery

Administrative-capacity research in CDBG-DR commonly reports a single point estimate from a single specification. We propose a six-item measurement-sensitivity audit protocol whose deliverable is a specification-curve dashboard, not a pass/fail verdict. Demonstrating on 543 local administering jurisdictions (primary) from HUD DRGR/QPR records (2003–2023) linked to QCEW NAICS 925110 staffing proxies (80.6% BLS-suppressed) and CDC/ATSDR vulnerability indicators: the capacity-timeliness coefficient is not stably identified. The dashboard spans positive, near-zero, and negative estimates depending on suppression handling, sample scope, capacity operationalization, and analytical framework; sign reversals and bootstrap confidence intervals crossing zero confirm the instability. A companion Cox survival analysis is itself measurement-sensitive (null under time-varying specification, positive under baseline). Capacity findings from this measurement stack are not robust enough for benchmarking without the protocol's sensitivity checks.

Evidence for Practice

Before comparing CDBG-DR jurisdictions on spending-rate or capacity metrics, practitioners and oversight agencies should apply a six-item measurement-sensitivity audit:

1. **Verify proxy suppression rates.** Before treating a staffing or capacity proxy as a benchmark, check its missing/zero/suppression rate in your sample and re-estimate on the non-suppressed subsample. If the coefficient changes materially, do not report the pooled result as a standalone benchmark. (Here: 80.6% of NAICS 925110 records are BLS-suppressed; the coefficient attenuates from 0.257 (local-only reference) to 0.132 n.s. on non-suppressed jurisdictions.)
2. **Test for maturity-driven sample selection.** If your analytic sample is constructed by complete-case selection on a duration outcome, run a maturity-stratified sensitivity or a fixed-horizon outcome. Mature-portfolio restrictions reversed the sign of the coefficient here ($\beta = -0.244$).
3. **Run at least one operationalization bridge.** Hold the analytical framework fixed and substitute an alternative capacity operationalization. Raw portfolio counts reversed the coefficient here ($\beta = -0.443$); financial-flow ratios reversed it further ($\beta = -0.600$).
4. **Compare across at least one alternative framework.** Report which design dimensions differ (unit, outcome, censoring, operationalization, covariates). A time-varying Cox on the same data returned null capacity-outcome associations here ($HR \approx 1.0$).
5. **Document covariate vintage and granularity.** For time-varying contextual covariates, document vintage and spatial granularity; flag sign reversals across vintages. SVI Theme 1 flips sign between 2010 and 2014+ vintages here.
6. **Use cluster-appropriate inference.** Compute standard errors under state- or disaster-clustered bootstrap rather than conventional ML SEs. The capacity-timeliness 95% CI crosses zero here under state-clustered bootstrap.

The protocol's deliverable is a **specification-curve dashboard** (§5.4), not a pass/fail verdict; readers apply their own fitness-for-use threshold. Statistical stability does not imply policy-readiness: direction-stable coefficients still require separate causal-identification arguments before anchoring oversight decisions.

2. Introduction

The effective administration of disaster recovery funds across affected jurisdictions is central to advancing equitable and timely community recovery. In the United States, the Community Development Block Grant-Disaster Recovery (CDBG-DR) program has become one of the largest and most flexible sources of long-term recovery assistance following major disaster declarations (Costa et al. 2026). Yet despite its national importance, the pace at which grantees—states, counties, cities, and territories—obligate, disburse, and expend CDBG-DR funds varies markedly across jurisdictions and disaster events. Slow or inconsistent financial performance has been repeatedly linked to delayed housing reconstruction, prolonged displacement, uneven economic recovery, and persistent inequities for low-income households and historically marginalized communities (Rouhanizadeh et al. 2020, 2022).

A key factor influencing financial performance and program outcomes is the administrative capacity of grantees—their institutional ability to move funds efficiently through the federal pipeline (Miao et al. 2025). However, capacity in disaster recovery is inherently difficult to measure directly. Prior CDBG-DR research has relied on single-period financial indicators (Cevik and Huang 2018; Dzigbede et al. 2020; Lindsay 2014), case-based assessments (Hamideh 2015; Rosas et al. 2021), or single-specification structural models. These designs share a common vulnerability: they typically report a single estimate from a single specification without systematic sensitivity to how capacity is operationalized, how complete-case selection on duration outcomes shapes the analytic sample, which analytical framework is adopted, or how within-cluster dependence affects inference. Earlier conceptualizations have framed administrative capacity as reflecting the combined effects of resources, workload manageability, and recovery outcomes (Kusumasari et al. 2010); raw financial measures alone are insufficient for characterizing the administrative-resource and workload-manageability dimensions that may shape those outcomes.

The contribution is to reframe capacity research in CDBG-DR as a measurement-sensitivity audit rather than as an attempt to recover a stable substantive effect. We propose a six-item audit protocol for administrative-capacity studies in this domain and demonstrate it on a national CDBG-DR dataset, showing that single-specification capacity findings can yield positive, null, or negative substantive conclusions on the same data depending on which audit items are exercised. The protocol’s value is to make those design dependencies visible and reproducible.

The research questions are therefore:

1. What minimum set of measurement-sensitivity checks should administrative-capacity studies in CDBG-DR report alongside any headline coefficient?
2. When that protocol is applied to a national CDBG-DR dataset, how much of the substantive conclusion changes across audit items?
3. What practical recommendations follow for benchmarking, oversight, and future research?

To address these questions, we assemble 573 administering-jurisdiction profiles (543 local, 30 state agency) from HUD DRGR/QPR records (2003–2023), linked to QCEW staffing/payroll proxies and CDC/ATSDR Social Vulnerability Index measures. The local-only SEM is the primary analytical specification because QCEW-based staffing measurement differs systematically between state and local populations. [Table 5.1](#) (§4.2) enumerates the nine designs that exercise the audit items; the dashboard in §5.4 reports the coefficient spread.

3. Literature Review

3.1 CDBG-DR Administration and the Measurement Problem

CDBG-DR funding has expanded substantially over the past two decades, allocating billions of dollars in response to major hurricanes, floods, wildfires, and other disasters (Costa et al. 2026). The program’s block-grant flexibility introduces administrative burden, and evaluations document persistent delays related to procurement, environmental reviews, and program start-up (Costa et al. 2026; Martin 2018; Miao et al. 2025; U.S. Department of Housing and Urban

Development 2024). Grantees vary substantially in prior experience, staffing, and procedures—factors that can influence timeliness and effectiveness of fund deployment (Martin et al. 2022; Meléndez 2020; Van Niekerk et al. 2018). Despite the program’s scale, relatively little research has systematically modeled grantees’ administrative capacity using DRGR/QPR records; prior studies typically rely on descriptive summaries or single-period financial indicators (Costa et al. 2026; Martin 2018).

3.2 Four Methodological Approaches and Their Sensitivity Exposures

Four broad approaches characterize the existing literature. *Descriptive and programmatic assessments* (HUD reports and GAO audits) document variation in fund utilization but rarely model institutional determinants (U.S. Government Accountability Office 2019; U.S. Department of Housing and Urban Development 2024). *Regression and panel-data analyses* examine how staffing, experience, governance, or severity influence implementation (Cutter et al. 2016; Lee 2019; Nor 2025); these studies typically report a single specification on observed financial indicators. *Case studies and qualitative work* identify recurring implementation barriers but do not generalize across jurisdictions (Jordan et al. 2014; Ospina et al. 2018; Rouhanizadeh et al. 2020). *Latent-variable methods*, particularly SEM, have been applied to disaster preparedness, resilience, and adaptive capacity (Geddami et al. 2024; Gendeshmin et al. 2025), though applications to CDBG-DR finance remain limited.

Each approach has measurement-sensitivity exposures rarely surfaced in print: descriptive work depends on the chosen ratio (disbursed/obligated vs. expended/disbursed); regression work on whether the analytic sample is constructed by complete-case selection on duration outcomes; SEM on whether the latent factor is sufficiently identified; case-based work on case selection. Capacity findings in this domain depend on staffing-proxy availability (typically QCEW NAICS 925110, which is heavily suppressed in small jurisdictions), portfolio maturity at observation, and how within-cluster dependence is treated in inference (Martin et al. 2019; Safapour et al. 2021; Peacock et al. 2014). The audit protocol developed in §4.1 makes those exposures auditable; the demonstration that follows applies the protocol to a national CDBG-DR dataset linking DRGR/QPR records, QCEW, population, and SVI indicators. Table 3.1 maps a selection of prior CDBG-DR and disaster-recovery capacity studies to the six audit items. The pattern is clear: no study in this sample addresses more than one item, and most report a single specification without any of the measurement-sensitivity checks the protocol prescribes.

Table 3.1: Prior CDBG-DR and disaster-recovery capacity studies mapped to the six audit items (§4.1). ✓ = addressed; ◦ = partially addressed (e.g., clustered SEs reported but not compared to conventional); — = not addressed. Assessment is based on published methodology descriptions and may undercount sensitivity reporting in studies that exercised audit items without foregrounding them in the methods section. The table illustrates the field’s typical reporting practice, not to adjudicate individual studies; a study coded as — may have addressed the item in supplementary materials not reviewed here.

Study	Approach	1. Proxy transp.	2. Sample select.	3. Within-bridge	4. Cross-framework	5. Vintage/g ran.	6. Cluster infer.
Costa	Descri	—	—	—	—	—	—

Study	Approach	1. Proxy transp.	2. Sample select.	3. Within-bridge	4. Cross-framework	5. Vintage/g ran.	6. Cluster infer.
et al. (2026)	ptive						
U.S. Government Accountability Office (2019)	Audit	—	—	—	—	—	—
U.S. Department of Housing and Urban Development (2024)	Programmatic	—	—	—	—	—	—
Martin (2018)	Descriptive	—	—	—	—	—	—
Martin et al. (2022)	Regression	—	—	—	—	—	◦
Miao et al. (2025)	Regression	—	—	—	—	—	—
Peacock et al. (2014)	Regression	—	—	—	◦	—	◦
Dzigbede et al. (2020)	Panel	—	—	—	—	—	—
Geddam et al. (2024)	SEM	—	—	—	—	—	—
This analysis	SEM + Cox	✓	✓	✓	✓	✓	✓

Study	Approach	1. Proxy transp.	2. Sample select.	3. Within-bridge	4. Cross-framework	5. Vintage/g ran.	6. Cluster infer.
s							

4. Data

4.1 Data Description and Sources

The QPR dataset was obtained from the U.S. Department of Housing and Urban Development’s DRGR reporting system and contains quarterly information on obligated, disbursed, and expended CDBG-DR funds for grants and reporting activities. The records span 18 disaster contexts from 2003 through 2023 and include thousands of organization labels that appear in grantee and activity fields (U.S. Department of Housing and Urban Development 2024). Quarterly records were used to construct study-window outcome indicators for each administering jurisdiction, including the disbursed-to-obligated ratio, the expended-to-disbursed ratio, the share of obligated funds fully expended, the proportion of programs completed, the duration of grant completion, and the average duration of program completion. Three terminology distinctions are important: *grantee* refers to HUD’s DRGR administrative label, *activity responsible organization* refers to the raw quarterly reporting name, and *administering jurisdiction* refers to the analytic SEM unit after those reporting entities are linked to a common state-agency or county-linked local-government profile. *Grant* denotes the disaster-specific CDBG-DR allocation, whereas *activity* denotes the underlying DRGR/QPR reporting unit nested within a grantee’s disaster portfolio. The quarter-varying spending indicators therefore summarize the observed reporting history of each administering jurisdiction rather than a single quarter snapshot.

Administrative resource data were drawn from the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW). Employment and payroll measures tied to NAICS 925110 (Administration of Housing Programs) are used as indirect proxies for disaster-recovery administrative resources (U.S. Bureau of Labor Statistics 2024). These measures provide a standardized national frame, but they should not be interpreted as direct counts of staff assigned specifically to CDBG-DR. Disaster recovery administration may be distributed across agencies, temporary teams, contract support, and cross-departmental arrangements that are only partially captured by this industry code. Notably, 85.1% of local jurisdictions report zero QCEW employment for this industry code, likely reflecting BLS suppression of small-cell counts rather than a literal absence of housing-program staff (see Appendix A.3 for distributions). The QCEW proxy therefore carries meaningful variation only among the 15% of local jurisdictions with non-suppressed data and among state agencies (all of which report non-zero values).

Contextual measures of social vulnerability were derived from the CDC/ATSDR Social Vulnerability Index (SVI) at the state and county levels. The analysis uses the four standard SVI themes—socioeconomic status, household composition and disability, minority status and language, and housing type and transportation—as broad contextual covariates, not as precise longitudinal trend measures (CDC/ATSDR Geospatial Research, Analysis, and Services

Program 2024). Because CDC warns against literal cross-vintage comparison of SVI percentiles without careful harmonization, these variables are interpreted as standardized contextual descriptors in the final SEM-ready dataset, not as year-by-year measures of change.

The same QPR data system also provides the foundation for the complementary survival analysis, which constructs a grantee-disaster-quarter panel (N = 151 grantee-disaster pairs) using standardized financial flow ratios as time-varying capacity indicators. This alternative analytic sample preserves within-grantee temporal variation instead of aggregating to a cross-sectional jurisdiction profile.

4.2 Data Cleaning and Sample Construction

Activity Responsible Organization labels were standardized and linked to Census geographies through hierarchical matching (county/parish names first, then city-to-county lookup (US Census Bureau 2023)); see Appendix A.5 for the match composition and [Table 9.5](#) for stage-by-stage counts. Of 3,147 distinct ARO labels in the raw QPR universe, 995 resolve to county/city geographies (capturing \$9.6B in obligated funds); 444 institutional entities (\$32.3B) and 1,708 unresolved labels (\$38.5B) are excluded. The SEM sample therefore covers the clearly geography-linked subset, not the full CDBG-DR spending universe. The final complete-case SEM sample contains 573 administering-jurisdiction profiles (30 state agencies matched on state abbreviations; 543 local jurisdictions matched on county/parish/borough labels against the 2023 Census county file). Records with fewer than four quarterly observations were excluded. [Table 9.1](#) (Appendix A.1) reports the full data flow; [Table 9.6](#) (Appendix A.5) reports characteristics of included and excluded ARO categories. Geography-dependent findings should be interpreted as informative cross-sectional summaries, not as exact administrative-boundary measures.

5. Methods

5.1 The Six-Item Measurement-Sensitivity Audit Protocol

The protocol specifies a minimum set of measurement-sensitivity checks that administrative-capacity studies in CDBG-DR should report alongside any headline coefficient. Each item names a design dimension along which substantive findings can shift, prescribes a check, and specifies what to report.

1. **Proxy transparency.** For any administrative or staffing proxy, report the missing/zero/suppression rate disaggregated by analytically relevant subgroups, and run a sensitivity restricted to the non-suppressed subsample. Report whether the headline coefficient attenuates, retains significance, or changes sign.
2. **Sample-selection sensitivity.** Where the analytic sample is constructed by complete-case selection on duration-based outcomes, report a maturity-stratified sensitivity *or* re-estimate using a fixed-horizon outcome that does not require observed completion. Disclose the share of the underlying universe the complete-case sample represents.
3. **Within-framework operationalization bridge.** Hold the analytical framework fixed and rerun with at least one alternative operationalization of the headline construct—for

“capacity,” candidates include raw counts vs. ratios, staffing-scaled vs. financial-flow indicators, observed composites vs. latent factors.

4. **Cross-framework triangulation.** Report at least one alternative analytical framework on the same data, cataloging which design dimensions differ (unit, outcome, censoring, operationalization, covariate set).
5. **Vintage and granularity documentation.** For time-varying contextual covariates, document vintage and spatial granularity. Rerun across vintages where multiple exist and report sign reversals.
6. **Cluster-appropriate inference.** Compute standard errors under cluster-appropriate inference (e.g., state- or disaster-clustered bootstrap) rather than conventional ML SEs that assume independence. Report whether the headline 95% CI crosses zero under cluster adjustment.

The protocol’s purpose is to make design dependencies visible and reproducible in print. The Results section that follows exercises each audit item on the CDBG-DR data assembled here.

The six items are sufficient to surface the instability documented in §5 and to distinguish same-estimand from alternative-estimand sensitivity across this measurement stack. The demonstration also surfaces additional design dimensions—temporal alignment of the staffing proxy to the outcome window (§4.3), geography-matching uncertainty (Appendix A.5), ratio-construction coupling between numerator and denominator (Appendix C.1), and cross-vintage covariate harmonization (Appendix C.9)—that are consequential but nest within or adjacent to the six items rather than constituting cleanly separable audit dimensions. The protocol is therefore presented as a **demonstrated minimum for studies combining suppressed industry-code proxies, duration-based outcomes, and latent-variable modeling**, with scope for extension to other measurement stacks.

5.1.1 Estimand Taxonomy and the Specification-Curve Dashboard

The audit produces a *specification-curve dashboard* in the spirit of specification-curve (Simonsohn et al. 2020) and multiverse (Stegen et al. 2016) analyses, not a pass/fail verdict: readers see the spread of defensible estimates across the six audit items, annotated by perturbation class and stability, and apply their own fitness-for-use criterion.

Dashboard rows are classified into five classes carrying different inferential weight. Class II is split into II-C (capacity-operationalization change) and II-O (outcome change) because these probe different dimensions of estimand sensitivity:

- **Class Ia — Measurement-preserving perturbations (same estimand, same population, same conditioning set).** Here: state-clustered bootstrap, weak-indicator drop, residualized burden, QCEW imputation bounds.
- **Class Ic — Conditioning-set changes (same estimand, same population, altered conditioning set).** Adding covariates changes the conditional expectation that the coefficient estimates; the estimand is preserved only under the assumption that the added covariates are not mediators. Here: maturity-band controls, portfolio-scale controls.
- **Class Ib — Sample-scope perturbations (same construct, different population).** Differences reflect either measurement instability or population heterogeneity. Here:

local-only, nonzero-QCEW, local-AND-nonzero-QCEW, mature-only, NDR/Mitigation-excluded.

- **Class II-C — Capacity-operationalization change (same outcome, reoperationalized capacity).** Here: raw-portfolio-counts bridge (workload volume rather than staffing-scaled burden); within-SEM financial-ratio bridge (flagged II-C* for construct reassignment since the same ratios serve as outcome indicators in the reference SEM; Appendix C.1).
- **Class II-O — Outcome change (same capacity, reoperationalized outcome).** Here: reconstructed-panel fixed-horizon regressions (expenditure share at quarters 8, 12, 16 replaces the duration-based latent Recovery Timeliness; Appendix C.8).
- **Class III — Simultaneous design-dimension changes.** Here: SEM vs. Cox (Audit Item 4).

Rows are annotated with three stability labels: *Stable* (sign retained, magnitude $\geq 50\%$ of reference), *Attenuated* (sign retained, magnitude $< 50\%$, or cluster CI crosses zero), *Reversed* (sign changes). These labels are **descriptive visual-aid annotations** for reading the dashboard, not formal decision rules or hypothesis tests; the 50%-magnitude threshold is an analyst convention chosen so that readers can see at a glance which perturbations halve the reference coefficient, and readers should apply their own fitness-for-use judgment rather than treating the labels as quasi-statistical verdicts.

5.2 Design Matrix

Table 5.1 summarizes the analytical designs reported in the main text and appendices, to help readers track what changes across specifications. Each row is a design that exercises one or more audit items.

Table 5.1: Design matrix across main-text and appendix analyses

Design	Unit	N	Outcome	Censoring	Capacity op.	Covariates
Reference SEM (§5.1–5.3)	Jurisdiction	573	Latent Performance + Timeliness	Complete-case	Staffing-scaled workload + QCEW	state_level, population, 4 SVI themes
Observed-variable regression (C.7)	Jurisdiction	573	6 observed outcome indicators	Complete-case	Same 4 capacity indicators	Same
Fixed-horizon regression, grantee level (C.8)	Grantee	32–36	Expenditure share at q = 8/12/16	Fixed horizon	Same 4 capacity indicators	Same
Fixed-	Jurisdiction	102–128	Expenditure	Fixed	Same 4	Same

Design	Unit	N	Outcome	Censoring	Capacity op.	Covariates
horizon regression , reconstructed panel (C.8)	n		re share at q = 8/12/16	horizon	capacity indicators	
Raw-portfolio-counts bridge (C.1 row 14)	Jurisdiction	573	Latent Performance + Timelines	Complete-case	log(programs) + log(disasters)	Same
Financial-ratio SEM bridge (C.1 row 15)	Jurisdiction	573	Latent Timelines	Complete-case	Disbursed /obligated + expended/ disbursed as capacity	Same
Time-varying Cox survival (§5.5)	Grantee-disaster pair	151	Time to 95% completion	Right-censored	Lagged (1q) disbursed/ obligated + expended/ disbursed	gov type, grant size, prior exp., year, population, severity, employment
Single-spell Cox threshold (C.5)	Grantee-disaster pair	69–110	Time to X% completion	Right-censored	Baseline (q = 3) financial ratios	Same as Cox
Cox staffing bridge (C.5)	Grantee-disaster pair	100	Time to 95% completion	Right-censored	Staffing-scaled workload as static	Same as Cox

The designs differ on up to five dimensions (unit of analysis, outcome specification, censoring treatment, capacity operationalization, covariate set); sample size N varies as a consequence of the unit-of-analysis choice and is reported per design. One-dimension-at-a-time bridges change exactly one dimension: the within-SEM bridges vary only capacity operationalization; the Cox staffing bridge varies the same within survival; the observed-variable regression varies only latent-vs-observed outcomes; the fixed-horizon regression varies only censoring-vs-fixed-

horizon outcomes. The reference SEM vs. time-varying Cox divergence varies all five simultaneously.

5.3 Structural Equation Modeling

5.3.1 Model Selection and Unit of Analysis

Structural equation modeling (SEM) is employed to develop latent constructs representing administrative capacity and disaster recovery outcomes and to evaluate the cross-sectional associations among them. The design does not estimate moderation effects. Instead, the state-versus-local indicator, jurisdiction size, and social vulnerability measures enter the SEM as observed covariates, and any state/local contrasts discussed later are treated as descriptive subset comparisons rather than as formal interaction tests. SEM remains useful in this setting as a structured way to summarize multiple indicators measured on different scales, but the pooled measurement model should be read as an exploratory approximation rather than as a demonstrated invariance result across state and local cases.

The analysis uses a cross-sectional dataset of 573 administering-jurisdiction profiles, including 30 state agencies and 543 local jurisdictions. Each observation summarizes an administering jurisdiction's resource and recovery profile across the study window rather than representing a single quarter or a single disaster episode. This aggregation permits a national SEM, but it is also a strong modeling assumption because it compresses variation across disaster cohorts, federal policy periods, staffing arrangements, and portfolio complexity into one cross-sectional profile.

5.3.2 Conceptualization of Administrative Capacity

“Administrative capacity” is used here as a working label for a set of administratively recorded indicators of resources and workload, not as a claim about an underlying generative construct. Existing literature on disaster-recovery administration treats capacity broadly as the institutional ability of public agencies to administer CDBG-DR funds under differing levels of workload and contextual constraint (Khalid and Ashraf 2024; El-Taliawi and Van Der Wal 2019), and this framing motivates the choice of indicators. The two-factor SEM specification used here is therefore best understood as an exploratory decomposition of those indicators into a resources dimension and a workload dimension for descriptive and associational purposes, not as the validation of a coherent latent trait. Administrative Resources are represented by staffing and payroll intensity proxies derived from QCEW-based employment and payroll measures, and Workload Manageability is represented by staff-scaled workload indicators that relate the number of disaster programs and disaster events to available staffing. The Workload Manageability latent factor appears in equation notation and in structural-table column headers as `AdminBurdenCapacity` for consistency with the SEM software output; the two labels refer to the same latent factor throughout. Recovery Performance and Recovery Timeliness are treated strictly as outcomes, not as additional components of capacity.

A temporal-framing caveat applies. The QCEW indicators are computed as the mean of annual averages across the years each jurisdiction had any active CDBG-DR grant (Appendix A.3) — *contemporaneous* with the recovery outcomes, not pre-existing or baseline capacity. Because the averaging window overlaps the outcome window, the proxy and the outcome share temporal scope: changes in CDBG-DR staffing that occur *during* recovery contribute to the “capacity”

measure that is being used to explain recovery outcomes. The proxy may therefore co-evolve with the workload and outcome measures it is being used to explain. The audit results in §5.4 should not be read as estimating an effect of pre-existing capacity on recovery timeliness; a baseline or lagged-year QCEW sensitivity is recommended as future work, though baseline-year values inherit the same suppression structure for the suppressed majority and would not resolve the suppression problem on their own.

A specification tension is worth noting explicitly: the indicators may be better understood as formative components that jointly constitute capacity rather than as reflective manifestations of a single latent trait. In a reflective specification, each indicator is caused by an underlying administrative dimension; in a formative specification, the indicators are determinants of a composite. The models reported here adopt a reflective specification for estimation convenience, but this choice is not validated and the boundary estimates observed in several indicators (factor loadings fixed at 1.000, zero residual variances) are consistent with the exploratory rather than confirmatory reading. The factor loadings should therefore be read as exploratory summaries of covariation among administratively recorded indicators, not as evidence that a single latent trait generates the observed indicators.

The headline estimand throughout the dashboard (§5.4) is the standardized structural path from Workload Manageability (AdminBurdenCapacity) to Recovery Timeliness: a specific associational claim that jurisdictions with more manageable workloads tend to implement recovery faster, conditional on the SEM's covariate set. The Panel B alternative-estimand rows in the dashboard—raw portfolio counts, financial-flow ratios, and the Cox framework—probe adjacent but different constructs and should not be read as sensitivity tests of the same underlying capacity concept; they are exploratory bridges documenting how the substantive conclusion changes when the construct itself is substituted.

The administrative resource indicators are operationalized as population-scaled QCEW measures. Let `avg_employment` denote mean annual QCEW employment in NAICS 925110 expressed as a rate per thousand population, `avg_payroll` denote the analogous payroll rate, and `E_TOTPOP` denote Census total population (in thousands):

$$\text{avg_employment} = \frac{\text{total_QCEW_employment}}{\text{E_TOTPOP}}$$

$$\text{avg_payroll} = \frac{\text{total_QCEW_payroll}}{\text{E_TOTPOP}}$$

Two additional indicators reflect administrative workload intensity relative to staffing levels. The reference specification uses a population-interaction denominator that rescales the per-capita employment rate back to jurisdiction-level staffing equivalents, so that workload-per-staff values are comparable across jurisdictions of widely different sizes:

$$\text{programs_per_staff} = \frac{\text{Num_Program}}{\text{avg_employment} \times \text{E_TOTPOP} + \varepsilon}$$

$$\text{disasters_per_staff} = \frac{\text{Num_Disaster}}{\text{avg_employment} \times \text{E_TOTPOP} + \varepsilon}$$

The epsilon offset is $\varepsilon = 10^{-6}$ to avoid division by zero when QCEW employment is suppressed. The choice of ε within the ε -offset family is analytically inconsequential: z-standardization removes the ε scaling, and Appendix C.10 reports an ε -sensitivity scan across $\varepsilon \in \{10^{-3}, 10^{-12}\}$ confirming invariance to numerical precision. What matters is the *treatment* of suppressed zeros as if they were literal zeros, which the QCEW imputation-bounds analysis (Appendix C.10) probes by substituting non-zero alternatives. Appendix A.3 documents the suppression distribution; Appendix C.1 reports the employment-only denominator ($\text{avg_employment} + \varepsilon$) as a sensitivity. Both workload measures are reverse-coded so that higher SEM input values correspond to lower relative burden.

5.3.3 Conceptualization of Disaster Recovery Outcomes

Disaster recovery outcomes are modeled as two SEM factors: Recovery Performance and Recovery Timeliness. Recovery Performance is an exploratory composite of financial-flow-through and completion indicators, not a validated latent construct; it is measured using four indicators: the disbursed-to-obligated ratio, the expended-to-disbursed ratio, the share of obligated funds fully expended, and the proportion of programs completed. Together, these measures summarize how effectively jurisdictions move funds through the recovery pipeline and convert those funds into completed programs. These are cumulative administrative summary measures, not event-history outcomes. Because DRGR portfolios are updated through corrections, reimbursements, and accounting adjustments, some observed cumulative ratios can exceed 1.0 in the raw administrative file, especially for expended-to-disbursed and fully-expended measures. The main SEM retains those observed values rather than mechanically truncating them; [Table 9.14](#) (Appendix C.4) reports a capped-ratio sensitivity check.

Recovery Timeliness represents the speed and consistency of program and grant implementation. It is measured using the duration of grant completion and the average duration of program completion, both reverse-coded so that higher values indicate faster implementation:

$$\text{rev_duration_of_grant_completion} = - \text{duration_of_grant_completion}$$

$$\text{rev_average_duration_program_completion} = - \text{average_duration_program_completion}$$

These are useful summary indicators, but they are also sensitive to maturity and right-censoring. Newer grants and ongoing portfolios have had less time to close out, so timeliness is treated as a cross-sectional summary of observed completion pace, not as a fully uncensored event-history outcome. The implications of this sensitivity are examined in the Robustness section and the Limitations.

5.3.4 Control Variables

To account for contextual differences across jurisdictions, the SEM includes governmental level, jurisdiction size, and social vulnerability covariates (Peacock et al. 2014; CDC/ATSDR Geospatial Research, Analysis, and Services Program 2024). Governmental level is represented by a binary indicator coded 1 for state agencies and 0 for local governments. This variable is modeled as a direct predictor and later used to organize descriptive subset comparisons; it is not a moderation term. Jurisdiction size is captured through total population, and the four SVI themes are included as broad contextual controls. Because the final SEM-ready dataset stores these covariates in standardized form, coefficients reflect cross-sectional associations within the

aggregated administering-jurisdiction sample, not temporal effects of changing population or SVI conditions.

5.3.5 SEM Modeling Framework

The reference SEM specification decomposes administrative capacity into two related latent dimensions—Administrative Resources and Workload Manageability (the latter appears as **AdminBurdenCapacity** in equation notation and structural tables, per the convention noted above)—while representing disaster recovery outcomes as two related latent constructs—Recovery Performance and Recovery Timeliness.

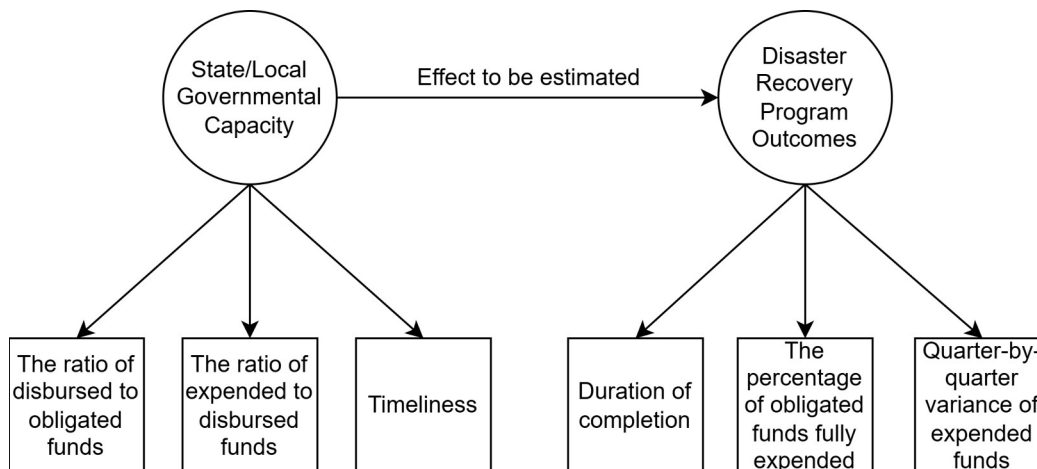


Figure 5.1: Structural equation model of administrative capacity and disaster recovery outcomes.

Both latent capacity dimensions are specified as predictors of Recovery Performance and Recovery Timeliness while controlling for governmental level, jurisdiction size, and the four SVI dimensions. The two capacity dimensions are allowed to covary, as are the two outcome constructs. All continuous indicators were standardized as z-scores prior to estimation, and the models were estimated with conventional maximum likelihood estimation (Lee and Zhu 2002).

An alternative one-factor SEM combining all four capacity indicators into a single factor (GovCapacity) was also estimated for comparison. Model evaluation relies on a combination of fit statistics—chi-square, CFI, TLI, RMSEA, AIC, and BIC—rather than on a single threshold. Model comparison is therefore treated as a tradeoff exercise: the two-factor model is substantively more informative and fits better on the main incremental and absolute fit measures, while the one-factor model remains more parsimonious by information criteria.

A series of robustness checks vary the specification (dropping weak indicators, residualizing burden indicators), add controls (maturity bands, portfolio scale), restrict the sample (cohort-restricted and stratified models), and test alternative burden indicators (raw portfolio counts instead of staffing-denominator ratios). These sensitivity analyses are reported in Appendix C and summarized in the main-text Robustness section. The complete-case SEM input contains 573 jurisdictions and 17 variables with no remaining missing values. Full model specifications are reported in Appendix B.

5.4 Complementary Survival Analysis

As a cross-framework complement to the SEM, a Cox proportional hazards model with time-varying covariates was estimated on the same QPR data system at the grantee-disaster level (Cox 1972). This alternative framework differs from the SEM simultaneously in five dimensions—unit of analysis, outcome specification, censoring treatment, capacity operationalization, and covariate set—exercising Audit Item 4.

First, the unit of analysis is the grantee-disaster pair ($N = 151$) rather than the administering-jurisdiction profile ($N = 573$). Second, the outcome is time to 95% expenditure completion, measured in quarters since first obligation. Third, right-censored observations are handled explicitly through the survival framework rather than incorporated as a duration indicator in a cross-sectional model; of the 151 grantee-disaster observations, 70 (46.4%) reached the 95% completion threshold within the observation window and 81 (53.6%) are right-censored. Smaller samples ($N = 142$ after government-type and size covariate drops; $N = 100$ for the staffing-indicator bridge after matching staffing data) appear in specific sensitivity analyses and are noted at each table. The `Duration_of_completion` outcome used by the SEM is defined as the number of months from first obligation to the quarter in which cumulative expended-to-obligated first reaches 0.95; portfolios that never reach 0.95 within the observation window have no observed `Duration_of_completion` and are excluded from the cross-sectional SEM through complete-case deletion on the timeliness indicators, whereas the survival framework retains them as right-censored observations. Fourth, capacity is measured through financial flow ratios (disbursed-to-obligated and expended-to-disbursed) as time-varying covariates with a one-quarter lag, rather than through staffing-scaled workload ratios in a cross-sectional profile. Fifth, the covariate set differs: the survival model includes static covariates for government type, grant size, prior CDBG-DR experience, disaster year, population, severity, and employment, compared to the SEM's `state_level`, population, and four SVI themes.

The 95% expenditure-completion threshold is an analytical approximation consistent with DRGR operational practice rather than a formal HUD statutory definition. DRGR/QPR portfolios routinely record small residual accounting adjustments in the final quarters (reimbursements, program-income netting, quality-assurance reviews) that cause cumulative expenditure ratios to hover just below 100% for months or years after substantive completion; treating the final 5% as accounting-reconciliation noise is a common analytical choice in administrative-data research on CDBG-DR, and [Table 9.15](#) (Appendix C.5) reports sensitivity across the 20%–100% range confirming the substantive time-varying finding does not hinge on this specific cutoff.

Bootstrap clustered standard errors (1,000 iterations, grantee-level clustering) account for within-grantee correlation. The model includes static covariates for government type, grant size, prior CDBG-DR experience, disaster year, population, severity, and employment. Full specification details are reported in Appendix B.5.

6. Results

The Results section presents the cross-sectional SEM first (§5.1–5.3) as the most specification-sensitive design and illustrative entry point into the dashboard, not as the preferred inferential

engine. The observed-variable OLS analogue (§5.3) produces a nearly identical coefficient (vs. $\beta_{\text{SEM}} = 0.257$), so readers concerned about the SEM’s latent-structure fragility can read the dashboard through the observed-variable lens without loss. Less-compromised designs for the substantive capacity-timeliness question — the non-suppressed-QCEW subsample (§5.4), the time-varying Cox (§5.5), and the fixed-horizon regression (Appendix C.8) — have smaller samples but do not inherit the ε -offset artifact or the complete-case maturity selection that distort the pooled SEM. The section exercises all six audit items across these designs.

6.1 SEM Illustrative Specification: Model Comparison

Both SEM specifications were estimated on the 573-jurisdiction complete-case sample. [Table 6.1](#) reports the comparison.

Table 6.1: SEM model fit comparison (N = 573).

Metric	One-Factor	Two-Factor
Chi-square (df)	741.72 (101)	469.82 (98)
Chi-square p-value	< 0.001	< 0.001
CFI	0.854	0.915
TLI	0.818	0.891
RMSEA	0.105	0.081
AIC	67.41	74.36
BIC	219.69	239.69

The model comparison reveals a genuine tradeoff rather than an unambiguous winner. The one-factor model has lower information criteria (AIC = 67.41; BIC = 219.69), but the two-factor model fits the covariance structure more convincingly on the conventional SEM diagnostics (CFI = 0.915 versus 0.854; TLI = 0.891 versus 0.818; RMSEA = 0.081 versus 0.105). The AIC/BIC preference for the one-factor model indicates that the two-factor choice is theoretically motivated rather than statistically compelled. Because the two-factor model aligns more closely with the distinction between resources and workload as separate dimensions of administrative capacity, and because it improves the main fit indices materially, it is retained as the illustrative specification for the audit demonstration—a worked example of how the protocol surfaces fragility—not as a privileged primary or validated latent structure.

The measurement parameters of the two-factor model motivate a cautious but interpretable reading. Within Administrative Resources, standardized loadings are strong for employment (0.953) and payroll (0.992). Within Workload Manageability, the reverse-coded programs-per-staff indicator loads at 0.577 and disasters-per-staff loads at approximately 1.000. Recovery Performance is less coherent: disbursed-to-obligated loads at 0.785, obligated funds fully expended at 0.988, programs completed at 0.477, and expended-to-disbursed at \$-0.131. Recovery Timeliness is identified by reverse-coded duration of completion (0.545) and reverse-coded average program duration (1.000). The two latent capacity dimensions are positively correlated (standardized covariance = 0.402), and the structural part of the model explains more variance in Recovery Timeliness than in Recovery Performance (approximate latent $R^2 = 0.20$).

versus 0.09). Full measurement parameters, structural paths, and latent covariances are reported in Appendix B.3 (Table 9.8, Table 9.9, and Table 9.10).

6.2 Structural Relationships Between Capacity and Recovery Outcomes

Table 6.2 reports standardized structural path coefficients. The **local-only sample (N = 543) is the primary analytical report** and pooled (N = 573) is supplementary: state-vs-local measurement invariance is not testable at the 30 state-agency cases, and the pooled model mixes two institutional populations with systematically different QCEW observability (100% non-suppressed for states; 15% for locals). Pooled results remain reported for comparison with prior literature.

*Table 6.2: Structural path estimates in the two-factor SEM — **local-only primary (N = 543); pooled supplementary (N = 573)**. The local-only model drops `state_level` (constant-0 on the subsample). Local-only fit: CFI = 0.902, TLI = 0.872, RMSEA = 0.092. Pooled fit: CFI = 0.915, TLI = 0.891, RMSEA = 0.081. Cluster-bootstrap 95% CIs (1,000 iter, state-level clustering; 33 clusters for local-only, 35 for pooled) for capacity paths only; full detail in Table 9.17 (Appendix C.6). Both local-only and pooled CIs for the headline capacity-timeliness path cross zero.*

Path	Local-only β (N=543)	Local-only cluster- bootstrap 95% CI	Pooled β (N=573)	Pooled cluster- bootstrap 95% CI	Local-only p	Pooled p
AdminResources → RecoveryPerformance	\$-0.170 [-\$0.320, \$0.009]	\$-0.170 [-\$0.319, \$0.004]	< 0.001	< 0.001		
AdminBurdenCapacity → RecoveryPerformance	\$+0.088 [-\$0.121, 0.249]	\$+0.077 [-\$0.139, 0.243]	0.053	0.086		
AdminResources → RecoveryTimeliness	\$+0.070 [-\$0.103, 0.243]	\$+0.080 [-\$0.083, 0.244]	0.118	0.072		
AdminBurdenCapacity → RecoveryTimeliness	\$+0.257 [-\$0.176, 0.542]	\$+0.266 [-\$0.129, 0.531]	< 0.001	< 0.001		
SVI Theme 3 (Minority/Language) → RecoveryPerformance	\$-\$0.204	—	\$-\$0.226	—	< 0.001	< 0.001

Path	Local-only β (N=543)	Local-only cluster-bootstrap 95% CI	Pooled β (N=573)	Pooled cluster-bootstrap 95% CI	Local-only p	Pooled p
mance						
SVI Theme 1 (Socioeconomic) → RecoveryTimeliness	\$-\$0.199	—	\$-\$0.194	—	0.004	0.004
SVI Theme 2 (HH Comp./Disability) → RecoveryTimeliness	\$-\$0.140	—	\$-\$0.159	—	0.015	0.004
State-vs-Local → RecoveryPerformance	(not modeled)	—	\$-\$0.038	—	—	0.505
State-vs-Local → RecoveryTimeliness	(not modeled)	—	\$-\$0.106	—	—	0.050

6.2.1 Recovery Performance and Timeliness paths

Two structural paths predict Recovery Performance with opposite signs: **AdminResources** → **Performance** is negative ($\beta = -0.170$, $p < 0.001$ in both samples; sign stability is tempered by the QCEW bounds analysis in §5.4 where suppression imputations flip the sign), while **AdminBurdenCapacity** → **Performance** is positive but marginal ($\beta = +0.088$ local-only, $p = 0.053$; $+0.077$ pooled). Treating them as a single “capacity effect” is incorrect.

Two paths predict Recovery Timeliness: **AdminBurdenCapacity** → **Timeliness** is positive ($\beta = +0.257$ local-only; bootstrap 95% CI [$-\$0.176, 0.542$] crossing zero; conventional ML $p < 0.001$ — the headline coefficient the audit in §5.4 perturbs), while **AdminResources** → **Timeliness** is positive but not significant ($\beta = +0.070$ local-only; bootstrap CI [$-\$0.103, 0.243$]; ML $p = 0.118$). The discrepancy between the conventional ML $p < 0.001$ and the bootstrap CI crossing zero for the headline path underscores the importance of Audit Item 6 (cluster-appropriate inference).

Among covariates, SVI Theme 3 (minority/language) is most clearly associated with Performance ($\beta = -0.204$ local-only, -0.226 pooled; $p < 0.001$) — read as contextual: jurisdictions serving more linguistically diverse or historically marginalized communities face greater outreach, verification, and service-delivery demands.

SVI Theme 1 (socioeconomic, $\beta = -0.199$, $p = 0.004$) and Theme 2 (household composition, $\beta = -0.140$, $p = 0.015$) associate with slower Timeliness in the local-only primary (pooled values $-\$0.194$ and $-\$0.159$).

These findings are vintage- and granularity-sensitive (Appendix C.9): state-level = -0.215) and 2014 – 2022 ($+\$0.167$ to $+\$0.460$); per-jurisdiction disaster-year re-estimation reassigns theme-to-outcome channels. Vulnerability findings require explicit vintage documentation; capacity coefficients are stable under SVI substitution but not under alternative QCEW suppression handling (§5.4, Appendix C.10).

The state-versus-local indicator is not meaningful for Performance and sits at the 0.05 threshold for Timeliness; governmental level is treated as a control, not a moderator.

Local-only and pooled conclusions match closely ($\beta = +0.257$ vs. $+0.266$ capacity-timeliness; -0.170 resources-performance in both), but similar coefficients demonstrate robustness to population composition, not latent-construct invariance (formal invariance testing is not feasible at $N = 30$ state cases). Both estimates share the ε -offset treatment of suppressed QCEW zeros; §5.4 develops the reading from departures from that treatment.

6.3 Audit Items 1, 2, 3, 6: Within-SEM Sensitivity

The reference capacity-timeliness association ($\beta = 0.257$, local-only) attenuates, stays stable, or reverses depending on the perturbation — the dashboard in [Figure 6.1](#) plots the specification curve across all rows, and [Table 6.3](#) reports the numerical values with audit-item class labels and stability flags; full details appear in Appendix C.

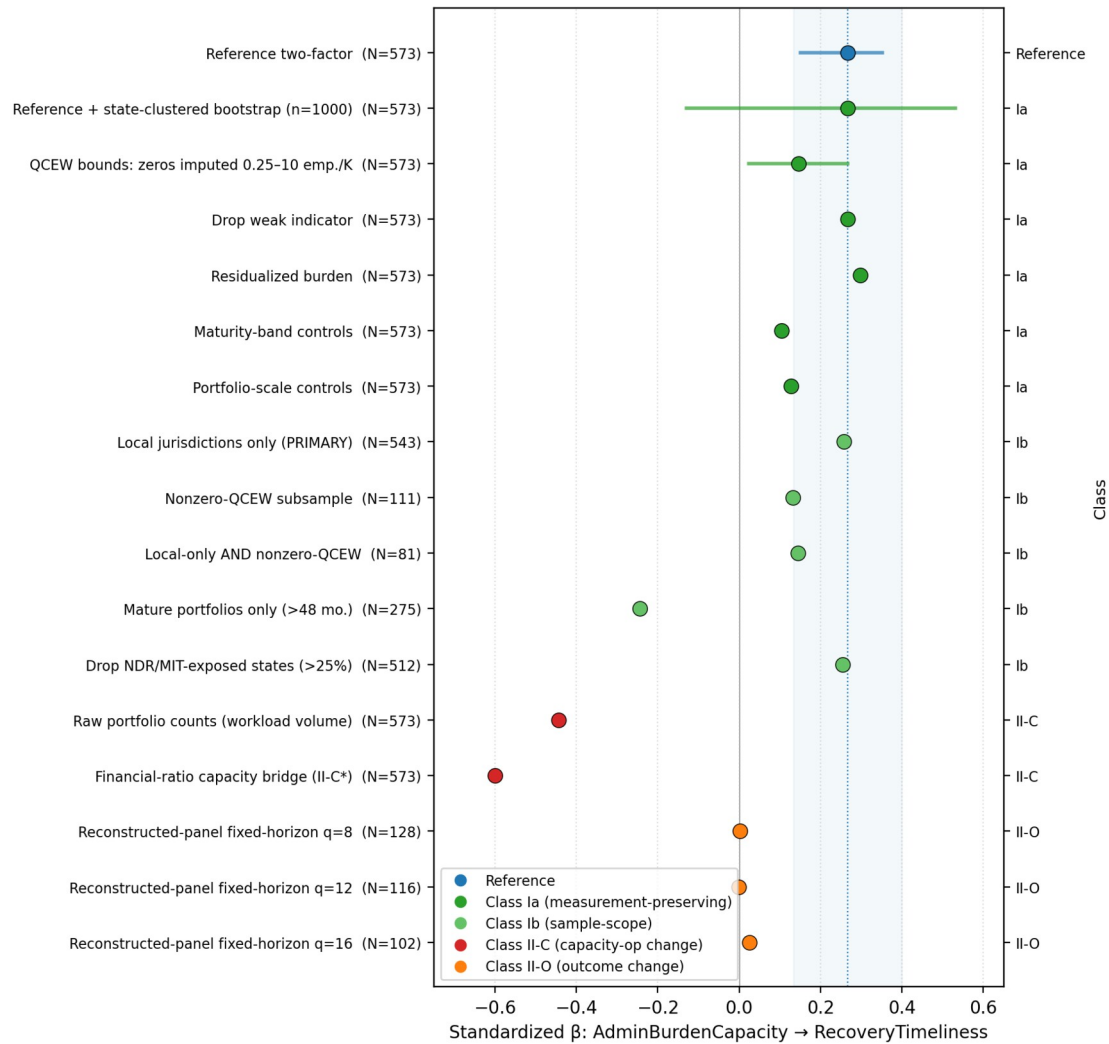


Figure 6.1: Specification-curve dashboard for the AdminBurdenCapacity → RecoveryTimeliness path across the perturbations enumerated in Table 6.3. Class Ia (measurement-preserving, dark green) and Ib (sample-scope, light green) perturb within the estimand; Class II-C (capacity-operationalization change, red) and Class II-O (outcome change, orange) change the estimand. The vertical dotted line marks the local-only reference $\beta = 0.257$; the shaded band marks the 50%-magnitude convention. The QCEW-bounds row shows the imputation range (0.25–10 emp./K pop). Class II-O rows (reconstructed-panel fixed-horizon at $q=8, 12, 16$) report a related but different estimand (expenditure share vs. duration-based latent Timeliness) and are shown for reference.

Table 6.3: Specification-curve dashboard for the capacity-timeliness path. The local-only two-factor SEM ($N=543$, $\beta = 0.257$) is the reference specification; pooled ($N=573$) is a Class Ib population-expansion perturbation. **Class Ia** = measurement-preserving within same estimand, population, and conditioning set; **Class Ic** = conditioning-set change (added

covariates may alter the conditional estimand); **Class Ib** = sample-scope perturbation; **Class II-C** = capacity-operationalization change (* denotes construct reassignment); **Class II-O** = outcome-operationalization change; **Class III** = simultaneous design-dimension changes. **Stability:** *Stable* = sign retained and magnitude $\geq 50\%$ of reference; *Attenuated* = sign retained, magnitude $< 50\%$, or cluster CI crosses zero, or not significant; *Reversed* = sign changes. Stability labels are descriptive annotations, not a pass/fail verdict. CFI is reported where available; rows with **CFI** < **0.90** (bolded) should be interpreted with caution, as poor model fit may contribute to coefficient instability independently of the measurement choice being tested.

Specific ation	Class	N	CFI	Burden → Timeliness β	Stability	Note
Panel A: Same-estiman d sensitivit y						
Local- only two- factor (referenc e)	—	543	0.902	0.257	—	Primary referenc e (local- only, conventi onal ML SE)
Local- only, state- clustered bootstra p (n=1000)	Ia	543	0.902	0.257, 95% CI [\$- 0.176, 0.542]] -\$0.129, 0.531] also crosses zero		
QCEW bounds: zeros imputed at 0.25– 10 emp./K pop	Ia	573	—	+0.024 to +0.267 (range)	Attenuat ed	Pooled frame; referenc e 0.266 at upper bound; Appendi x C.10
Drop weak indicator	Ia	573	0.919	0.266	Stable	Pooled frame

Specific ation	Class	N	CFI	Burden → Timeliness β	Stability	Note
Residual ized burden	Ia	573	0.918	0.297	Stable	Survives arithmeti c- coupling removal; pooled frame
Maturity -band controls	Ic	573	0.915	0.105	Attenuat ed	Below 50% of referenc e; conditio ning-set change
Portfolio -scale controls	Ic	573	0.783	0.127	Attenuat ed	Below 50%; conditio ning-set change; poor fit
Nonzero -QCEW subsamp le	Ib	111	0.918	0.132 (n.s.)	Attenuat ed	Non- suppress ed subsamp le; below 50% magnitu de
Local- only AND nonzero- QCEW	Ib	81	0.898	0.145 (n.s.)	Attenuat ed	Decomp oses state/loc al confoun d
Mature portfolio s only (>48 mo.)	Ib	275	0.829	-\$0.244	Reverse d	Sign reversal under maturity restrictio n; below-

Specific ation	Class	N	CFI	Burden → Timeliness β	Stability	Note
						threshold fit
Drop NDR/MI T- exposed states (>25%)	Ib	512	—	0.255	Stable	Non- event- allocatio n exposure not driving
Panel B: Alternati ve- estiman d triangul ation						
Raw portfolio counts (workloa d volume)	II-C	573	0.854	-\$0.443	Reverse d	Capacity -op change only
Financia l-ratio capacity bridge	II-C*	573	0.926	-\$0.600	Reverse d	Capacity -op change <i>plus</i> construct reassign ment (Append ix C.1)
Reconstr ucted- panel fixed- horizon q=8	II-O	128	—	+0.003 (n.s.)	Attenuat ed	Outcome changed to expendit ure share; Appendi x C.8
Reconstr	II-O	116	—	-\$0.001	Attenuat	Outcome

Specific ation	Class	N	CFI	Burden → Timeliness β	Stability	Note
ucted- panel fixed- horizon q=12				(n.s.)	ed	change
Reconstr ucted- panel fixed- horizon q=16	II-O	102	—	+0.025 (n.s.)	Attenuat ed	Outcome change
Time- varying Cox survival (Item 4)	III	151	—	HR $\approx$ 1.0 (null)	—	Multiple design dimensio ns differ

Reading the dashboard: the reference specification is local-only ($N = 543$, $\beta = 0.257$), matching the primary analytical specification throughout the paper. The local-only state-clustered bootstrap CI $[-\$0.176, 0.542]$ *crosses zero* (33 clusters); the pooled analogue ($= 0.266$, CI $[-\$0.129, 0.531]$) also crosses zero. Among **Class Ia** measurement-preserving perturbations on the pooled frame, two rows are stable (weak-indicator drop, residualization); the **QCEW imputation-bounds analysis** (Appendix C.10) yields a β range of $+0.024$ to $+0.267$. The two **Class Ic** conditioning-set additions—maturity-band ($\beta = 0.105$) and portfolio-scale ($\beta = 0.127$) controls—attenuate below 50% of reference; these are separated from Class Ia because adding covariates changes the conditioning set and arguably the estimand, not only the measurement. Among **Class Ib** sample-scope perturbations, the pooled model is stable ($\beta = 0.266$), NDR/Mitigation-excluded is stable ($\beta = 0.255$), two attenuate and lose significance (nonzero-QCEW $\beta = 0.132$, local-AND-nonzero-QCEW $\beta = 0.145$), and one reverses (mature-only $\beta = -0.244$).

The **Class II-C** capacity-operationalization bridges reverse sign: the raw-portfolio-counts bridge ($\beta = -0.443$, no construct contamination) and the financial-ratio bridge ($\beta = -0.600$, II-C, with construct reassignment per Appendix C.1). The **Class II-O** reconstructed-panel fixed-horizon regression (Appendix C.8, $N = 102$ –128) substitutes cumulative expenditure share at quarter 8, 12, or 16 for the SEM's duration-based latent Timeliness; because the outcome is different, these coefficients do not directly adjudicate the SEM estimate but report whether a related estimand shows a capacity association. On these data the II-O coefficient is $\beta \approx 0$ at every horizon. The **Class III** SEM vs. Cox comparison (§5.5) varies multiple design dimensions simultaneously. The dashboard is presented in two panels. **Panel A** (same-estimand sensitivity: Class Ia, Ic, and Ib) reports the spread of estimates that probe the same* capacity-timeliness association under different measurement, conditioning-set, and sample choices; the core instability claim rests on this panel. **Panel B** (alternative-estimand triangulation: Class II-C, II-O, III) reports estimates

that intentionally change the capacity construct, the outcome, or the entire analytical framework; these are informative about adjacent questions but are not directly comparable to the reference estimand. Across both panels — positive estimates, near-zero, sign reversals — the capacity-timeliness estimate is not stably identified under these measurement choices.

A commensurability caveat applies to the Panel A rows: because standardized structural betas from modified measurement models are not automatically on the same scale — the beta’s standardization depends on the latent factor’s variance, which shifts when indicators are added, dropped, or replaced — the dashboard rows are most reliably compared within the Class Ia subset where the factor structure is held constant. The observed-variable regressions in §5.3 and the fixed-horizon analyses (Appendix C.8), where scale and interpretation are transparent, provide a complementary reading that does not depend on latent-construct commensurability; the core instability claim is supported by both.

6.3.1 Audit Item 3 Specifically: Observed-Composite Bridge

The observed-variable analogue clarifies what the latent-construct machinery contributes.

Table 6.4 reports OLS of each reverse-coded timeliness indicator on the four capacity indicators with the full covariate set (Appendix C.7 reports the full six-outcome table).

Table 6.4: Observed-composite OLS for timeliness indicators ($N = 573$; z-scored). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Outcome (reverse-coded)	$b(\text{rev Programs/Staf f})$	$b(\text{rev Disasters/Staf f})$	$b(\text{Avg Employment})$	$b(\text{Avg Payroll})$	R^2
Duration of Completion	\$+\$0.276 ***	\$-\$0.099 .	\$+\$0.299 *	\$-\$0.183	0.094
Average Program Duration	\$+\$0.128 **	\$+\$0.187 ***	\$+\$0.100	\$-\$0.009	0.209

The observed coefficient on reverse-coded programs-per-staff ($b = +0.276$ on Duration of Completion) is nearly identical to the latent-construct SEM coefficient ($\beta_{\text{SEM}} = 0.257$ local-only, 0.266 pooled), so the SEM is functioning primarily as dimension reduction, not as latent-construct validation. The latent-variable machinery is therefore dispensable for the headline substantive conclusion: the measurement-sensitivity verdict holds whether the analysis is conducted as an SEM or as an observed-variable regression.

The QCEW proxy underlying Audit Item 1 behaves in two distinctive ways on the SEM sample. Figure 6.2 panel (a) shows the zero-rate breakdown: 0% of state agencies, 85.1% of local jurisdictions, 80.6% of the pooled sample have zero QCEW employment in NAICS 925110. Panel (b) shows that the reference burden-indicator denominator ($\text{avg_employment} \times E_TOTPOP + \epsilon$) has a strongly bimodal distribution: one mode near the $\epsilon = 10^{-6}$ floor (most local jurisdictions, where the denominator collapses to ϵ and the staffing-scaled ratio reduces to a

raw program/disaster count scaled by 10^6) and a second mode at substantively meaningful values (state agencies and the 15% of locals with non-suppressed QCEW). The reference $\beta = 0.266$ is therefore a weighted mixture of two regimes: a majority regime in which the “staffing-scaled” ratio is effectively a raw count, and a minority regime (the nonzero-QCEW subsample of 111) in which it behaves as the construct describes.

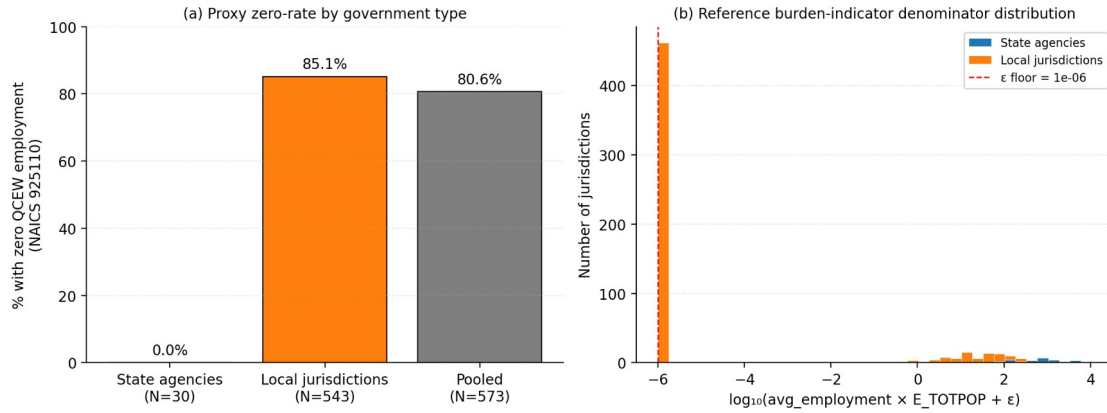


Figure 6.2: QCEW proxy behavior on the 573-jurisdiction SEM sample. (a) Share of jurisdictions reporting zero QCEW employment in NAICS 925110 by government type. (b) Distribution of the reference burden-indicator denominator ($\text{avg_employment} \times E_TOTPOP + \epsilon$, $\epsilon = 10^{-6}$) on a $\log_{10}$ scale, stacked by government type. The left mode near $\log_{10} \epsilon = -6$ is the population of jurisdictions where the denominator collapses to the epsilon floor.

6.4 Audit Item 4: Cross-Framework Triangulation

The Cox proportional hazards model with time-varying covariates at the grantee-disaster level yields substantively different conclusions from the cross-sectional SEM. Table 6.5 reports the key results.

Table 6.5: Cox proportional hazards model with time-varying covariates ($N = 151$ grantee-disaster pairs). Bootstrap clustered standard errors (1,000 iterations, grantee-level clustering). Model concordance = 0.723. Full coefficient table, proportional-hazards residual tests, and threshold-sensitivity results across the 20%–100% completion range are reported in Appendix B.5 and Appendix C.5.

Capacity Measure	Hazard Ratio	95% CI	p-value
Disbursement ratio (lagged)	1.001	[0.821, 1.221]	0.991
Expenditure ratio (lagged)	0.998	[0.728, 1.367]	0.990

Neither lagged financial flow ratio is associated with program completion timing in the time-varying specification. The hazard ratios are effectively 1.0; concordance of 0.723 indicates the covariates discriminate completion timing through other channels than the lagged ratios.

The Cox framework also exhibits within-framework measurement sensitivity. A complementary single-spell Cox specification with the same financial ratios measured at baseline (Appendix C.5) reports positive and significant hazard ratios on baseline disbursement (HR = 1.46 at the 20% threshold rising to HR = 2.58 at 100%; $p < 0.01$ throughout), though estimates at the highest thresholds ($\geq 90\%$) rest on small event counts (20–7 events) and should be read with caution. The time-varying and single-spell specifications ask different questions on the same data (quarterly change vs. baseline level) and reach different answers: a within-Cox divergence parallel to the within-SEM Class II bridge reversals.

The cross-framework evidence therefore supports a narrower conclusion than a clean SEM-vs-Cox contrast would: analytically reasonable designs yield different answers on the same data, and the within-Cox instability means some of what looks like framework-level divergence is actually within-framework measurement sensitivity. The cross-framework comparison does not yet identify which design dimension drives the SEM-vs-Cox gap. Single-method studies should be interpreted with caution regardless of framework.

7. Discussion

7.1 What the Audit Protocol Surfaces

The capacity-timeliness coefficient is not stably identified under these measurement choices. The dashboard spans positive (local-only reference $\beta = 0.257$; pooled 0.266; residualized 0.297), near-zero (non-suppressed 0.132 n.s.; reconstructed-panel fixed-horizon ≈ 0 ; time-varying Cox HR ≈ 1.0), and negative (mature-only -0.244 ; raw counts -0.443 ; financial-ratio bridge -0.600) estimates. Plausible design choices yield substantively different conclusions on the same data; capacity findings built on this measurement stack—suppressed QCEW NAICS 925110 staffing proxies, duration-based complete-case selection, ε -offset SEM estimation, and pooled state-plus-local aggregation—are therefore not robust enough for benchmarking or policy anchoring. Studies using a different proxy, censoring-aware outcomes, or censored-data estimators may reach different conclusions; the indeterminacy documented here is a property of this particular design combination, not a universal claim about CDBG-DR capacity.

Among the measurement-appropriate Class Ia and Ic specifications, the coefficient clusters near zero — the QCEW imputation-bounds range is $+0.024$ to $+0.267$, with the pooled ε -offset reference at the *upper* bound. This near-zero central tendency on the measurement-appropriate subset is consistent with a reading in which the pooled magnitude is largely produced by the ε -offset treatment of BLS-suppressed zeros. The reading does not extend to the Class Ib sample-scope results (mature-only sign reversal) or the Class II-C bridges (raw counts, financial-ratio); the overall dashboard supports indeterminacy, not a near-zero verdict. QCEW suppression is a first-order measurement problem; full resolution would require a censored-data estimator not available in semopy.

The two headline capacity paths behave differently under suppression-handling alternatives. The burden $\rightarrow$ timeliness coefficient remains positive but attenuates sharply across the imputation scan (from $+0.267$ to $+0.024$), suggesting direction stability conditional on the measurement stack but not magnitude stability. The resources $\rightarrow$ performance coefficient ($\beta = -0.170$ under ε -offset)

reverses sign entirely to +0.249 at QCEW imputation = 10 (BLS-suppression-consistent), so this path is direction-unstable across principled suppression-handling alternatives.

7.2 Social Vulnerability and Recovery Outcomes

Contextual vulnerability matters across the specifications reported here: the directional pattern that more vulnerable communities experience slower or less complete recovery is more stable than the capacity-timeliness coefficient. However, the specific SVI theme-to-outcome assignments (e.g., minority/language → Performance, socioeconomic → Timeliness) are vintage- and granularity-sensitive (Audit Item 5; Appendix C.9) and should be treated as hypotheses for design-appropriate evaluation rather than as identified mechanisms.

7.3 What the Cross-Framework Design Differences Add

Audit Item 4 (cross-framework triangulation, §5.5) is the item most often absent from single-method CDBG-DR research. The SEM-vs-Cox comparison varies five design dimensions simultaneously (unit, outcome, censoring, operationalization, covariates), so the divergence should be read as evidence that design choices matter collectively, not as one framework invalidating the other. Two within-framework bridges narrow the candidates: a Cox model with staffing covariates returns null hazard ratios, and the within-SEM financial-ratio bridge reverses sign ($\beta = -0.600$). The within-Cox instability (time-varying null vs. baseline positive) reported in §5.5 adds a further caution: some of what looks like cross-framework disagreement is within-framework measurement sensitivity.

7.4 Practical Implications

The protocol translates into operational guidance for two audiences. *Researchers* publishing administrative-capacity studies get a reproducible minimum reporting standard. *Oversight agencies* using DRGR/QPR-derived metrics get evidence that single-specification capacity-outcome ratios can support substantively opposite conclusions, so spending-rate benchmarks should be accompanied at minimum by proxy-zero-rate, nonzero-subsample, and operationalization-bridge checks (audit items 1 and 3). Social vulnerability is stable in direction (vulnerable communities experience slower or less-complete recovery) but the specific theme-to-outcome assignments are vintage-sensitive (audit item 5); vulnerability-informed targeting requires explicit vintage documentation.

A concrete example illustrates the planning consequence. Under the ϵ -offset specification, 307 of 573 jurisdictions rank above the sample median on Workload Manageability; 196 of those 307 (64%) have BLS-suppressed QCEW employment, so their above-median ranking reflects having few programs in the portfolio rather than adequate staffing. Under the non-suppressed subsample, 462 jurisdictions (81%) become unclassifiable entirely, and 55 of the 111 remaining jurisdictions that were above-median under ϵ -offset fall below-median when only jurisdictions with genuine staffing data are compared. An oversight agency that benchmarks jurisdiction capacity using the pooled ϵ -offset ranking would therefore be benchmarking portfolio size for most of the sample; applying Audit Item 1 (proxy transparency) before the benchmark makes that dependence visible.

8. Limitations

The audit protocol surfaces several measurement-sensitivity exposures in the demonstration data; these are properties of the available CDBG-DR records as much as limitations of the analytic designs reported here.

First, the demonstration is associational across all designs. The reference SEM aggregates quarterly QPR data to one profile per administering jurisdiction, discarding within-jurisdiction variation across disasters, policy regimes, staffing arrangements, and maturation windows; the Cox specification preserves time variation but does not identify causal effects. The protocol is intended to characterize how design choices change associational estimates, not to recover a causal parameter.

Second, the QCEW NAICS 925110 staffing proxy used by Audit Item 1 is heavily suppressed (85.1% zero among local jurisdictions; 0% among state agencies) and does not directly measure CDBG-DR administrative teams, contractors, or interagency support structures. The Administrative Resources factor in the SEM may therefore partly track level of government rather than within-type administrative capacity. Formal measurement-invariance testing between state and local groups is not feasible in this sample because $N = 30$ state cases is insufficient for stable parameter estimation. The audit makes this exposure visible (Item 1 nonzero-subsample sensitivity) rather than resolving it.

Third, the duration-based timeliness indicators are sensitive to right-censoring and disaster maturity, and the SEM sample is constructed through complete-case deletion on those indicators, biasing it toward more-mature and more-complete portfolios. The Audit Item 2 fixed-horizon regressions in Appendix C.8 partially address this exposure at grantee level ($N = 32\text{--}36$) and at the F31-reconstructed jurisdiction level ($N = 102\text{--}128$), but neither reaches the full $N = 573$ frame. Rebuilding the panel around fixed-horizon outcomes at scale is the most consequential design improvement available for this data system.

Fourth, the cross-framework comparison (Audit Item 4) varies multiple design elements simultaneously and is informative about framework sensitivity in the aggregate but is not a one-dimension-at-a-time test of which channel drives the SEM–Cox divergence; the within-framework bridges reported in §6.3 narrow but do not exhaust the candidates.

Fifth, the geography layer for the final SEM sample is drawn from administrative profiles that resolved cleanly to Census GEOIDs; profiles requiring manual adjudication upstream that did not survive complete-case exclusion are not represented. Audit Item 6 cluster-appropriate inference uses a 1,000-iteration state-clustered bootstrap (Appendix C.6); multiway clustering at the SEM unit is not naturally definable because each jurisdiction-profile aggregates multiple disaster contexts.

Sixth, the protocol has scope conditions: it is built for studies combining suppressed industry-code proxies, complete-case selection on duration outcomes, latent-variable modeling on small samples, and clustered observational structure. Domains without these features may need to add or substitute items; the six reported here are a minimum, not a maximum.

These exposures point to a research agenda for the data system itself: longitudinal or event-history designs with fixed-horizon outcomes, one-dimension-at-a-time bridges across SEM and

survival frameworks, and more direct administrative-process data—staffing assignments, contractor use, application processing times, intergovernmental coordination mechanisms—that could move analyses from proxy capacity measures toward direct operational indicators of recovery governance.

9. Conclusion

The contribution is a six-item measurement-sensitivity audit protocol whose deliverable is a specification-curve dashboard, demonstrated on 543 local administering jurisdictions (primary), 30 state agencies (supplementary), and 151 grantee-disaster pairs. Applied to these data, the capacity-timeliness coefficient is not stably identified: the local-only reference yields $\beta = +0.257$; measurement-appropriate specifications cluster near zero; sample-scope and operationalization perturbations produce sign reversals (-0.244 to -0.600). The survival framework is itself measurement-sensitive: null under time-varying specification, positive and significant under baseline. The burden $\rightarrow$ timeliness path attenuates sharply under suppression-handling alternatives but retains its positive sign; the resources $\rightarrow$ performance path reverses sign entirely. Capacity findings built on this measurement stack—suppressed QCEW proxies, duration-based complete-case outcomes, ϵ -offset SEM, pooled aggregation—are not robust enough for benchmarking or policy anchoring; a different measurement stack may support different conclusions, and surfacing that dependence is the protocol’s point.

The protocol is portable. Its items and dashboard framing are not specific to CDBG-DR; they apply to administrative-capacity research combining heavily suppressed industry-code staffing proxies, complete-case selection on duration outcomes, latent-variable modeling on small samples, and clustered observational structure. Where those ingredients co-occur, the audit surfaces that single-specification capacity findings are unlikely to survive principled perturbation. Improving disaster recovery governance (Tierney 2012) will require routine measurement-sensitivity audits and, where possible, better direct measurement than sectoral QCEW proxies allow.

Acknowledgements

Partially supported by the U.S. Department of Housing and Urban Development (Award# H21749CA). The findings and conclusions are those of the authors and do not represent the official position or views of the U.S. Department of Housing and Urban Development.

Disclosure Statement

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. Generative AI tools were used only for limited language editing. All analytic decisions, interpretation, and manuscript content were reviewed and approved by the authors.

Data Availability Statement

Reproducibility falls into three tiers. **(1) Fully reproducible from public data:** HUD CDBG-DR DRGR/QPR reports, U.S. BLS QCEW files (NAICS 925110), and CDC/ATSDR SVI (vintages 2010–2022) are publicly available; all SEM estimation, bootstrap, survival-model, and figure-generation scripts are included in the replication package. **(2) Reproducible conditional on the deposited crosswalk:** the 573-jurisdiction crosswalk mapping each Grantee label to state_level, state abbreviation, and inferred geography type is deposited with the replication package so that downstream analyses can be reproduced without repeating the upstream geography linkage. **(3) Not independently recoverable:** the manual adjudication decisions that resolved ambiguous Activity Responsible Organization labels in the upstream data-assembly workflow (approximately 1,708 unresolved ARO labels excluded from the final SEM sample) are documented in the replication log but cannot be mechanically reconstructed by an independent team. See Appendix A.5 for the match composition of the 573 surviving profiles.

References

CDC/ATSDR Geospatial Research, Analysis, and Services Program. 2024. *CDC/ATSDR Social Vulnerability Index (SVI) Data and Documentation: Vintages 2000, 2010, 2014, 2016, 2018, 2020, 2022*. <https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html>.

Cevik, Serhan, and Grace Huang. 2018. *How to Manage the Fiscal Costs of Natural Disasters*. {IMF} How-To Note 18/03. International Monetary Fund.

Collett, David. 2015. *Modelling Survival Data in Medical Research*. 3rd ed. Chapman; Hall/CRC.

Costa, Ricardo, Brian Mann, Atefeh Sobhani, Sara Hamideh, Ali Nejat, and Ashley Ross. 2026. “From Allocation to Action: A Comparative Analysis of CDBG-DR Funding Expenditures for 2017 Disasters in California, Florida, and Texas.” Unpublished manuscript.

Cox, David R. 1972. “Regression Models and Life-Tables.” *Journal of the Royal Statistical Society, Series B (Methodological)* 34 (2): 187–220.

Cutter, Susan L., Kevin D. Ash, and Christopher T. Emrich. 2016. “Urban-Rural Differences in Disaster Resilience.” *Annals of the American Association of Geographers* 106 (6): 1236–52.

Dzigbede, Komla D., Sarah B. Gehl, and Katherine Willoughby. 2020. “Disaster Resiliency of U.S. Local Governments: Insights to Strengthen Local Response and Recovery from the COVID-19 Pandemic.” *Public Administration Review* 80 (4): 634–43.

El-Taliawi, Ola G., and Zeger Van Der Wal. 2019. “Developing Administrative Capacity: An Agenda for Research and Practice.” *Policy Design and Practice* 2 (3): 243–57.

Geddam, Sheshadri Mohan, Pradeep Kumar Garg, and Anil Kumar Tyagi. 2024. “Enhancing Disaster Management Effectiveness: A Structural Equation Modeling Approach.” *International Journal of Disaster Risk Reduction* 99: 104123.

- Gendeshmin, Saeed Bagherzadeh, Siavash Karami, and Sayed Mohammad Mousavi. 2025. "Structural Equation Modeling for Disaster Risk Assessment: A Systematic Review." *International Journal of Disaster Risk Reduction* 119: 105301.
- Hamideh, Sara. 2015. "Planning for Disaster Recovery: Lessons from Hurricane Ike for Theory and Practice." PhD thesis, Texas A&M University.
- Jordan, Elizabeth, Amy Javernick-Will, and Bernard Amadei. 2014. "A Qualitative Comparative Analysis of Neighborhood Recovery Following Hurricane Katrina." *International Journal of Disaster Resilience in the Built Environment* 5 (4): 391–412.
- Khalid, Saeeda, and Iram Ashraf. 2024. "The Role of Administrative Capacity in Disaster Management." *Social Trends and Policy Review* 2 (2): 97–133.
- Kronmal, Richard A. 1993. "Spurious Correlation and the Fallacy of the Ratio Standard Revisited." *Journal of the Royal Statistical Society, Series A (Statistics in Society)* 156 (3): 379–92.
- Kusumasari, Bevaola, Quamrul Alam, and Kamal Siddiqui. 2010. "Resource Capability for Local Government in Managing Disaster." *Disaster Prevention and Management* 19 (4): 438–51.
- Lee, Dong Wook. 2019. "Local Government's Disaster Management Capacity and Disaster Resilience." *Local Government Studies* 45 (6): 803–26.
- Lee, Sik-Yum, and Hong-Tu Zhu. 2002. "Maximum Likelihood Estimation of Nonlinear Structural Equation Models." *Psychometrika* 67 (2): 189–210.
- Lindsay, Bruce R. 2014. *FEMA's Disaster Relief Fund: Overview and Selected Issues*. {CRS} Report R43537. Congressional Research Service.
- Martin, Carlos. 2018. *The Evidence Base on How CDBG-DR Works for State and Local Stakeholders*. Urban Institute.
- Martin, Carlos, Brett Gilbert, Daniel Teles, and Brett Theodos. 2019. *Housing Recovery and CDBG-DR*. Urban Institute.
- Martin, Carlos, Daniel Teles, and Nicole DuBois. 2022. "Understanding the Pace of HUD's Disaster Housing Recovery Efforts." *Housing Policy Debate* 32 (1): 102–27.
- Meléndez, Edwin. 2020. "Puerto Rico Housing and Community Development Industry's Capacity for Disaster Recovery." *Centro Journal* 32 (3): 83–118.
- Miao, Qing, Katherine Best, Allison Reilly, and Meri Davlasheridze. 2025. "Where Did the Money Go? Assessing the Distribution of the Community Development Block Grant-Disaster Recovery Funds." *The American Review of Public Administration* 55 (3): 227–48.
- Nor, Mohd Idris. 2025. "Investigating the Dynamics of Climate Finance Disbursements: A Panel Data Approach from 2003 to 2022." *PLOS One* 20 (3): e0318170.

- Ospina, Sonia M., Marc Esteve, and Seoyoun Lee. 2018. "Assessing Qualitative Studies in Public Administration Research." *Public Administration Review* 78 (4): 593–605.
- Peacock, Walter Gillis, Shannon Van Zandt, Yang Zhang, and Wesley E. Highfield. 2014. "Inequities in Long-Term Housing Recovery After Disasters." *Journal of the American Planning Association* 80 (4): 356–71.
- Rosas, Estelina, Patrick S. Roberts, Andrew Lauand, Itzel A. Gutierrez, and Blas Nunez-Neto. 2021. "Assessing the Impact of Municipal Government Capacity on Recovery from Hurricane Maria in Puerto Rico." *International Journal of Disaster Risk Reduction* 61: 102340.
- Rouhanizadeh, Bahareh, Sharareh Kermanshachi, and Tara Jahan Nipa. 2020. "Exploratory Analysis of Barriers to Effective Post-Disaster Recovery." *International Journal of Disaster Risk Reduction* 50: 101735.
- Rouhanizadeh, Bahareh, Sharareh Kermanshachi, and Elnaz Safapour. 2022. "An Investigation of Causal Relationships Among Barriers That Delay Post-Hurricane Recovery of Human Communities." *International Journal of Disaster Risk Reduction* 67: 102666.
- Safapour, Elnaz, Sharareh Kermanshachi, and Apurva Pamidimukkala. 2021. "Post-Disaster Recovery in Urban and Rural Communities: Challenges and Strategies." *International Journal of Disaster Risk Reduction* 64: 102535.
- Simonsohn, Uri, Joseph P. Simmons, and Leif D. Nelson. 2020. "Specification Curve Analysis." *Nature Human Behaviour* 4 (11): 1208–14. <https://doi.org/10.1038/s41562-020-0912-z>.
- Steege, Sara, Francis Tuerlinckx, Andrew Gelman, and Wolf Vanpaemel. 2016. "Increasing Transparency Through a Multiverse Analysis." *Perspectives on Psychological Science* 11 (5): 702–12. <https://doi.org/10.1177/1745691616658637>.
- Tierney, Kathleen. 2012. "Disaster Governance: Social, Political, and Economic Dimensions." *Annual Review of Environment and Resources* 37: 341–63.
- U.S. Bureau of Labor Statistics. 2024. *Quarterly Census of Employment and Wages*. <https://www.bls.gov/cew/downloadable-data-files.htm>.
- U.S. Department of Housing and Urban Development. 2024. *CDBG-DR Reports and Quarterly Performance Reports*. <https://www.hud.gov/hud-partners/community-cdbg-dr-reports>.
- U.S. Government Accountability Office. 2019. *Disaster Recovery: Better Monitoring of Block Grant Funds Is Needed*. Report to Congressional Addressees GAO-19-232. GAO.
- US Census Bureau. 2023. *Incorporated Places (City-Level Geographies) and ANSI Codes*. Data-processing reference for city-to-county linkage; <https://www.census.gov/library/reference/code-lists/ansi.html>.
- Van Niekerk, Dewald, Leandri D. Nema-konde, Lukhanyo Kruger, and Kobus Forbes-Genade. 2018. "Community-Based Disaster Risk Management." In *Handbook of Disaster Research*, 2nd ed., edited by Havidán Rodríguez, William Donner, and Joseph E. Trainor. Springer.

Appendix A: Data Description

This appendix documents the analytic pipeline from raw administrative records to the 573-jurisdiction SEM sample, variable definitions, proxy distributions, and geography matching. The four data sources (HUD DRGR/QPR, BLS QCEW for NAICS 925110, CDC/ATSDR SVI, and 2023 Census county/state shapefiles) are described in the main-text Data section and not repeated here.

9.1 A.1 Data Flow

Table 9.1 summarizes the pipeline from raw administrative records to the 573-jurisdiction SEM sample.

Table 9.1: Data flow from raw QPR to analytic sample

Stage	Count	Description
Standardized DRGR/QPR activity-quarter rows	130,605	Repository-standardized administrative rows
Rows with valid actual-quarter labels	128,382	Excludes rows with missing QPR Actual Quarter
Collapsed grantee-disaster-quarter rows	3,623	Quarter-collapsed standardized panel
Recovered archived jurisdiction profiles	577	Raw bundled CSV before exclusions
Reference complete-case SEM sample	573	30 state agencies + 543 county-linked local jurisdictions

The four excluded profiles are grantees with missing values on one or more SEM indicators after z-score standardization. The 573-jurisdiction sample represents all administering jurisdictions with complete data on the 10 SEM indicator variables and 6 covariates.

9.2 A.2 Variable Dictionary

Table 9.2 defines all variables used in the SEM.

Table 9.2: Variable definitions

Variable	Construction	Source	Direction
avg_employment	Mean QCEW annual employment (NAICS 925110), expressed as a rate per thousand population	BLS QCEW	Higher = more staff per capita

Variable	Construction	Source	Direction
avg_payroll	Mean QCEW annual payroll (NAICS 925110), expressed as payroll per thousand population	BLS QCEW	Higher = more payroll per capita
programs_per_staff	$\text{Num_Program} / (\text{avg_employment} \times \text{E_TOTPOP} + \varepsilon)$; denominator rescales per-capita employment to jurisdiction-level staffing equivalents	Derived	Higher = heavier workload
disasters_per_staff	$\text{Num_Disaster} / (\text{avg_employment} \times \text{E_TOTPOP} + \varepsilon)$; denominator as above	Derived	Higher = heavier workload
Ratio_disbursed_to_obligated	Cumulative disbursed / cumulative obligated	DRGR/QPR	Higher = better flow-through
Ratio_expended_to_disbursed	Cumulative expended / cumulative disbursed	DRGR/QPR	Higher = better utilization
Ratio_obligated_funds_fully_expended	Share of obligated funds reaching full expenditure	DRGR/QPR	Higher = more completion
Ratio_Program_Completed	Share of programs reaching completion	DRGR/QPR	Higher = more completion
Duration_of_completion	Months from first obligation to 95% expenditure	DRGR/QPR	Higher = slower
Average_Duration_Program_Completion	Mean program-level completion duration	DRGR/QPR	Higher = slower
state_level	Binary: 1 = state agency, 0 = local jurisdiction	Derived	—
E_TOTPOP	Census total population (in thousands)	Census	—
SPL_THEME1	SVI: Socioeconomic vulnerability	CDC/ATSDR	Higher = more vulnerable

Variable	Construction	Source	Direction
SPL_THEME2	SVI: Household composition and disability	CDC/ATSDR	Higher = more vulnerable
SPL_THEME3	SVI: Minority status and language	CDC/ATSDR	Higher = more vulnerable
SPL_THEME4	SVI: Housing type and transportation	CDC/ATSDR	Higher = more vulnerable

All continuous variables are z-scored before SEM estimation. The burden indicators (programs_per_staff, disasters_per_staff) and duration indicators are reverse-coded so that higher values on all SEM indicators point in the conceptually positive direction (more capacity, better performance, faster timeliness).

9.3 A.3 QCEW Proxy Documentation

QCEW data are extracted for NAICS 925110 (“Administration of Housing Programs, Urban Planning, and Community Development”) at the county and state levels. This is an indirect proxy for CDBG-DR administrative capacity: it captures employment and payroll in the broader housing-program administration sector, not disaster-recovery staff specifically. It omits contractors, temporary teams, interagency staffing, and disaster-specific surge capacity.

Table 9.3 reports pre-standardization distributions by government type. The most important finding is that 85.1% of local jurisdictions report zero QCEW employment, likely reflecting BLS suppression of small-cell counts rather than a literal absence of housing-program staff. This floor-value concentration means the QCEW proxy carries meaningful variation only among the 15% of local jurisdictions with non-suppressed data and among state agencies (all of which report non-zero values).

Table 9.3: QCEW proxy distributions before standardization

Variable	Group	N	Min	P25	Median	P75	Max	% Zero
avg_employment	All	573	0.00	0.00	0.00	0.00	10.50	80.6%
avg_employment	State	30	0.11	0.40	1.20	2.27	5.37	0.0%
avg_employment	Local	543	0.00	0.00	0.00	0.00	10.50	85.1%

Variable	Group	N	Min	P25	Median	P75	Max	% Zero
avg_payroll	All	573	0	0	0	0	129,361	80.6%
avg_payroll	State	30	2,692	6,591	21,950	29,861	112,612	0.0%
avg_payroll	Local	543	0	0	0	0	129,361	85.1%

The temporal alignment of QCEW data to the recovery outcome window deserves explicit description. For each administering jurisdiction, the `avg_employment` and `avg_payroll` indicators are computed as the unweighted mean of QCEW annual averages across the years in which that jurisdiction had any active CDBG-DR grant in the DRGR/QPR system between 2003 and 2023. They are not quarter-matched to QPR reporting periods, not weighted by allocation size, and not lagged relative to the outcome quarter. As a result, the staffing proxy is best read as a cross-sectional descriptor of mean staffing intensity in the housing-program-administration sector during the jurisdiction’s CDBG-DR administration window, rather than as a time-varying or pre-period instrument for capacity. A jurisdiction whose QCEW employment was high in years when it administered a 2005 disaster but low when it administered a 2017 disaster would receive an averaged value somewhere between the two; a within-jurisdiction temporal-disaggregation of QCEW would require a panel SEM design and is not implemented in the cross-sectional reference.

Three terms referring to “zero QCEW values” should not be conflated. (1) A **literal zero** would mean a jurisdiction is known to have no NAICS 925110 employment during the reporting window; QCEW’s suppression rules do not distinguish this case from the suppressed case. (2) A **BLS-suppressed value** is a cell where BLS has replaced the observed count with zero in the public-use file because the underlying count is below the disclosure-avoidance threshold (typically three or fewer establishments or an identifiability risk for a single establishment). For the 80.6% of the sample with zero reported values, both interpretations are possible, and without access to the confidential micro-data the two cannot be distinguished. (3) A **non-suppressed observation** is a cell where the reported value is non-zero or where BLS has documented zero employment. Throughout the main text and this appendix, “zero” refers to the observed public-use value and carries the suppression ambiguity; “non-suppressed subsample” refers to the 111 jurisdictions with reported non-zero values, which are necessarily free of the suppression ambiguity.

9.3.1 Transportability of the non-suppressed subsample

Whether the non-suppressed subsample is *transportable* to the broader population — i.e., whether substantive findings on those 111 (or 114 in the 577-row pre-complete-case frame) jurisdictions generalize to the 462 (or 463) zero-QCEW jurisdictions — depends on whether the two groups differ systematically on observable characteristics. They do, substantially. [Table 9.4](#) reports the comparison; data are at `data_work/diagnostics/zero_nonzero_qcew_comparison.csv`.

Table 9.4: Comparison of zero-QCEW (N=463) and non-suppressed-QCEW (N=114) jurisdictions in the 577-row pre-complete-case frame. Cohen's d standardized mean difference; Welch's t-test p-values (unequal variances). Counts are 462/111 in the 573-row complete-case frame.

Variable	Zero-QCEW mean	Non-suppressed mean	Cohen's <i>d</i>	Welch <i>p</i>
Population (thousands)	6.7	262.7	\$+\$0.65	< 0.001
State- agency fraction	0.000	0.272	\$+\$0.86	< 0.001
Number of CDBG-DR programs	14.5	74.4	\$+\$0.43	0.002
Number of disaster contexts	1.6	2.0	\$+\$0.30	0.010
Duration of completion (months)	51.8	51.0	\$-\$0.02	0.83
Avg program- completion duration (months)	38.9	26.8	\$-\$0.57	< 0.001
Disbursed/ obligated ratio	0.51	0.39	\$-\$0.48	< 0.001
Expended/ disbursed ratio	0.89	1.25	\$+\$0.24	0.07
Fully- expended share	0.79	0.58	\$-\$0.65	< 0.001
Program- completion share	0.95	0.84	\$-\$0.56	< 0.001
SVI Theme 1 (socioeconomic)	2.46	2.36	\$-\$0.12	0.25
SVI Theme	2.23	2.10	\$-\$0.31	0.003

Variable	Zero-QCEW mean	Non-suppressed mean	Cohen's <i>d</i>	Welch <i>p</i>
2 (household composition)				
SVI Theme	0.91	1.15	+\$0.63	< 0.001
3 (minority/la nguage)				
SVI Theme	2.66	2.85	+\$0.35	< 0.001
4 (housing/tra nsportation)				

The non-suppressed subsample is systematically larger (population 40× the zero-QCEW mean), more institutional (27% state agencies vs. 0%), administers larger portfolios (5× as many programs), takes less time per program (median program-completion duration 27 vs. 39 months), but completes a smaller share of obligated funds in full and a smaller share of programs to completion (0.58 vs. 0.79 fully-expended; 0.84 vs. 0.95 completion). It also serves communities with notably different vulnerability profiles (Theme 3 minority/language +0.63; Theme 4 housing/transport +0.35). The non-suppressed jurisdictions are therefore not a representative sample of the broader CDBG-DR-administering population — they are a biased subset of larger, busier, more institutional administrators with measurably different outcome profiles. Substantive findings restricted to this subsample (e.g., the $\beta = 0.132$ n.s. capacity-timeliness coefficient on $N=111$) describe the capacity association in this specific population segment, not the broader population. The transportability question — whether the same association would hold if the suppressed-majority were observable — cannot be answered without access to confidential QCEW micro-data or substantively richer staffing measurement.

9.4 A.4 SVI Vintage Specification

The reference SEM uses a single SVI vintage assigned to each jurisdiction at the midpoint of its CDBG-DR administration period, with the four theme-level scores (Socioeconomic; Household Composition/Disability; Minority Status/Language; Housing Type/Transportation) as covariates rather than the overall SVI composite. Because CDC/ATSDR warns against cross-vintage comparison of percentile scores, full vintage-sensitivity and per-jurisdiction disaster-year SVI analyses are reported in Appendix C.9.

9.5 A.5 Geography Matching

Jurisdictions in the DRGR/QPR system are identified by grantee labels that do not always correspond to standard Census geographic units. The matching procedure uses a three-step hierarchy:

1. **State agencies:** Two-letter state abbreviations matched directly to the 2023 Census state file (STUSPS field). Result for the SEM sample: 30/30 matched (100%).

2. **Local jurisdictions:** Organization names parsed to extract city/county identifiers, then matched to the 2023 Census county file by normalized county/parish/borough label and state. Result for the SEM sample: 543/543 matched (100%).
3. **Unresolved cases:** 0 SEM-sample jurisdiction labels could not be resolved to an official Census GEOID.

The 100% match rate above refers to the *surviving* SEM sample after complete-case deletion. The upstream universe is substantially larger and not all raw organization labels resolve cleanly. [Table 9.5](#) reports a reconstruction of the upstream pipeline using a dedicated linker script (`scripts/rebuild_upstream_sample_flow.py`) that reads the raw F31 file and classifies each distinct Activity Responsible Org label by match stage against the 2022 CDC/ATSDR county reference. This reconstruction produces exact counts at each *decidable* stage (direct county match, city-of-type label requiring crosswalk, institutional label, unresolved) but does not recover the original Kaifa pipeline’s manual adjudication table, which is not preserved in the repository.

Table 9.5: Upstream geography-matching pipeline counts, reconstructed from `data_work/diagnostics/upstream_sample_flow.csv`. Stages 1–6 are exact from the reconstruction; stage 7 (577) and stage 8 (573) are inherited from the recovered Kaifa pipeline. The reconstruction differs from the original upstream aggregation because manual adjudication of city-of-type and institutional labels to enclosing counties cannot be exactly reproduced without the Kaifa pipeline’s lookup tables. The 677 city-of-type labels and some fraction of the 1,708 unresolved labels were aggregated upstream to county-level or state-agency profiles, producing the 577 recovered jurisdictions.

Stage	Count	Note
Distinct Activity Responsible Org labels in raw F31	3,147	Raw QPR organization field
State-agency grantees identifiable from Grantee column	34	Grantee = 2-letter state code or full state name; 30 survive SEM complete-case deletion
Direct Census county-name match (normalized, suffix-aware)	318	County / Parish / Borough / Census Area / Municipio labels matched to 2022 SVI county reference
City-of-type labels requiring crosswalk	677	“City of X”, “Town of X”, “X, ST” — aggregated to enclosing county in upstream pipeline
Institutional / non-geographic labels	444	Universities, authorities, non-profits, LLC/Inc.,

Stage	Count	Note
Unresolved by automated rules	1,708	consortia, housing agencies Specific city/town/organization names that did not match county or institutional patterns
Aggregated to administering-jurisdiction profiles (recovered)	577	Upstream Kaifa pipeline output after city-to-county aggregation and manual adjudication
Surviving complete-case SEM sample	573	After dropping 4 profiles with missing SEM indicators

County-linked local units inherit county-level GEOIDs; state-administered units inherit statewide GEOIDs. Jurisdictions whose Activity Responsible Org labels did not resolve to a county or state profile upstream are *not* represented in the SEM sample. This selection should be read as part of Audit Item 2 (sample-selection sensitivity): a jurisdiction whose labels were too ambiguous to link is also a jurisdiction whose recovery outcomes are not analyzed here. The upstream-sample-flow reconstruction at `scripts/rebuild_upstream_sample_flow.py` provides the counts at each decidable stage; full exact reproduction of the original pipeline’s manual adjudication is documented as reconstructible only from the raw F31 file plus Census reference data, not from artifacts preserved in the code repository.

9.5.1 Reproducibility Boundary

See the three-tier Data Availability Statement in the main text for a summary. Script paths for the fully reproducible tier: linker reconstruction (`scripts/rebuild_upstream_sample_flow.py`), state-clustered bootstrap (`scripts/run_state_clustered_bootstrap_sem.py`, 1,000 iterations), local-only bootstrap (same script, `--sample local`), NDR/Mitigation exclusion (`scripts/run_ndr_mitigation_sensitivity_sem.py`), local-only SEM (`scripts/run_local_only_sem.py`), fixed-horizon regressions (Appendix C.8), QCEW imputation bounds (`scripts/run_qcew_bounds_sensitivity.py`), ϵ -sensitivity scan (`scripts/run_epsilon_sensitivity.py`), and figure generation (`scripts/make_audit_figures.py`). The 573-jurisdiction crosswalk is deposited as `data_work/replication/jurisdiction_crosswalk.csv`. Future revisions would benefit from re-running the upstream aggregation with an open-source city-to-county crosswalk (e.g., Census Bureau’s ANSI place codes) to extend the fully reproducible tier.

9.5.2 Included vs. excluded ARO characteristics

Table 9.6 reports approximate characteristics of the F31 Activity Responsible Org labels grouped by linker-stage classification. *Included* AROs are those that resolve directly to a Census county or are city-of-type labels that the upstream pipeline aggregated to enclosing counties (Stages 3

and 4 of [Table 9.5](#)). *Excluded — institutional* labels are universities, authorities, non-profits, LLCs, consortia, and similar entities (Stage 5). *Excluded — unresolved* are specific city/town/organization names that did not match either category (Stage 6).

Table 9.6: Approximate F31 ARO characteristics by linker-stage classification. Source: data_work/diagnostics/included_vs_excluded_comparison.csv.

Category	N AROs	Median rows/ARO	Median obligated/AR O () <i>Mean oblig</i>)	Total obligated (\$B)	
Included (county or city, Stages 3– 4)	995	10	1.0M	9.6M	9.6
Excluded — institution al (Stage 5)	444	13	3.7M	72.7M	32.3
Excluded — unresolve d (Stage 6)	1,708	10	1.0M	22.5M	38.5

Two caveats apply to this comparison. First, the F31 *aggregate dollar volumes* in the excluded categories are deceptively large because many institutional and unresolved AROs are aggregated upstream into the 30 state-agency profiles in the SEM sample (e.g., a state housing finance authority appears in F31 under its institutional name but flows into the state-agency SEM profile for that state). The SEM sample therefore captures more recovery-spending volume than the raw count of 995 included AROs would suggest. Second, the unresolved category is heterogeneous: some labels are small towns and cities that map cleanly to counties under a richer crosswalk (and would join the SEM sample under a future revision), while others are non-geographic entities (community-action agencies, transit authorities, university research consortia) that do not map to a single jurisdiction at all. The SEM sample is therefore best characterized as covering local jurisdictions whose F31 reporting labels resolve cleanly to a county or city geography under automated rules, plus state agencies; it does not include subsets of CDBG-DR administration that flow through non-geographic institutional entities.

9.6 A.6 Disaster Contexts

The 18 disaster contexts represented in the HUD CDBG-DR quarterly reporting are enumerated in [Table 9.7](#), covering disasters from 2001 through 2023 plus two HUD program-level allocation

categories (2013 National Disaster Resilience and 2015-2018 Mitigation) that are not disaster-specific in the strict sense but appear as distinct appropriations in the DRGR system.

Table 9.7: 18 disaster contexts in the CDBG-DR analytic sample

Disaster Context	Event Year	Census Vintage	Program Allocation?	FEMA Disaster Numbers
2001 World Trade Center	2001	2000	No	1391
2005 Hurricanes Katrina, Rita, Wilma	2005	2000	No	1602, 1603, 1604, 1606, 1607
2008 Hurricane Ike and Other Events	2008	2000	No	1780, 1791, 1794
2008 Midwest Floods	2008	2000	No	1763, 1771, 1773
2010 Severe Storms and Flooding	2010	2010	No	1935, 1924, 1927
2011 Multiple Disasters	2011	2010	No	1991, 1994, 4012, 4024
2011-2013 Hurricane Sandy and Other Events	2012	2010	No	4085, 4086, 4087
2013 National Disaster Resilience	2013	2010	Yes	(program allocation)
2015 Hurricane Joaquin, Patricia, and Other Events	2015	2010	No	4241, 4245

Disaster Context	Event Year	Census Vintage	Program Allocation?	FEMA Disaster Numbers
2015-2018 Mitigation	2016	2010	Yes	(program allocation)
2016 Louisiana Floods and Other Events	2016	2010	No	4277, 4280
2017 Hurricanes Harvey, Irma, and Maria	2017	2010	No	4332, 4336, 4339
2018 Disasters	2018	2010	No	4393, 4396, 4399, 4407
2019 Disasters	2019	2010	No	4420, 4466, 4485, 4473
2020 Hurricanes Laura/Delta/Zeta and 2021 Hurricane Ida and Wildfires	2020–2021	2020	No	(multiple)
2022 Disasters including Hurricanes Fiona and Ian	2022	2020	No	4671, 4673, 4675
Electrical Power Systems in Puerto Rico and the U.S. Virgin Islands	2017	2010	No	4339
Storms, Flooding, and Other Disasters in	2023	2020	No	4699, 4683

Disaster Context	Event Year	Census Vintage	Program Allocation?	FEMA Disaster Numbers
California, Alabama, and Georgia				

Inclusion criteria: any CDBG-DR allocation reported in the DRGR/QPR system that produced at least one quarterly performance report within the 2003–2023 observation window. Two entries (2013 National Disaster Resilience and 2015–2018 Mitigation) are program-level allocations rather than disaster-event responses and are retained in the analysis because they appear as distinct grantee-disaster rows in the administrative file. Exclusion criteria: allocations with no quarterly performance reports or those fully administered outside the 2003–2023 window.

9.7 A.7 Survival Sample Composition

The survival analysis operates on a grantee-disaster-quarter panel derived from the same QPR data system as the SEM. The authoritative sample size is $N = 151$ unique grantee-disaster pairs, observed over a quarterly time grid from each pair's first obligation through the end of the 2023 observation window. Of these 151 pairs, 70 (46.4%) reached the 95% completion threshold within the observation window and 81 (53.6%) are right-censored.

Smaller sample sizes appear in specific sensitivity analyses:

- **$N = 142$** reflects pairs with complete static-covariate data (government type, grant size, prior experience, disaster year, population, severity, employment). The 9-pair attrition is primarily grantees whose disaster-year or prior-experience covariate is missing.
- **$N = 100$** reflects pairs for which staffing indicators (`programs_per_staff` and `disasters_per_staff` from the cross-sectional SEM data) can be merged on the grantee label, used only in the Cox staffing-bridge analysis reported in Appendix C.5.

9.8 A.8 Duration_of_completion Computation

The `Duration_of_completion` variable used as a Recovery Timeliness indicator in the SEM is defined at the grantee-disaster-profile level as follows:

1. For each grantee-disaster pair, locate the earliest quarter with positive obligated funds and call that quarter q_0 .
2. Compute the cumulative expended-to-obligated ratio in each subsequent quarter q_t .
3. `Duration_of_completion` = $(q_t - q_0)$ expressed in months, where q_t is the first quarter with ratio ≥ 0.95 .
4. For grantee-disaster pairs that never reach the 0.95 ratio within the observation window, `Duration_of_completion` is missing (NaN). These pairs are excluded from the cross-sectional SEM through complete-case deletion on the Recovery Timeliness indicators; they are retained in the survival analysis as right-censored observations.

The analogous `Average_Duration_Program_Completion` is computed as the mean of program-level completion durations within each grantee-disaster profile, again with missing

values for pairs whose programs have not all reached completion. The cross-sectional SEM sample of 573 administering jurisdictions is therefore biased toward more-mature portfolios whose programs have had sufficient observation time to reach the 95% threshold. This maturity selection is the mechanism underlying the sign reversal documented in Appendix C.1 (mature-portfolios-only row of [Table 9.11](#)) and is the reason the survival framework, which handles right-censoring explicitly, is reported as a companion rather than as a robustness check.

Appendix B: Methodological Details

9.9 B.1 Model Specifications

Two SEM specifications are compared, both estimated on the 573-jurisdiction complete-case sample.

One-factor model. A single latent capacity factor loads on all four capacity-related indicators:

$$\text{GovCapacity} = \sim z_avg_employment + z_avg_payroll + z_rev_programs_per_staff + z_rev_disasters_per_staff$$

$$\text{RecoveryPerformance} = \sim z_Ratio_disbursed + z_Ratio_expended + z_Ratio_fully_expended + z_Ratio_Program$$

$$\text{RecoveryTimeliness} = \sim z_rev_Duration + z_rev_Average_Duration$$

$$\text{RecoveryPerformance} \sim \text{GovCapacity} + \text{covariates}$$

$$\text{RecoveryTimeliness} \sim \text{GovCapacity} + \text{covariates}$$

Two-factor model (reference). The four capacity indicators are split into two latent factors:

$$\text{AdminResources} = \sim z_avg_employment + z_avg_payroll$$

$$\text{AdminBurdenCapacity} = \sim z_rev_programs_per_staff + z_rev_disasters_per_staff$$

The outcome measurement model and structural regressions are the same as the one-factor model, except that both AdminResources and AdminBurdenCapacity enter as predictors. The two capacity factors are allowed to covary freely. The two outcome factors (RecoveryPerformance and RecoveryTimeliness) are also allowed to covary.

Covariates in both models: state_level (binary), z_E_TOTPOP, z_SPL_THEME1 through z_SPL_THEME4.

9.10 B.2 Estimation Details

Both models are estimated via maximum likelihood using the semopy library in Python. All continuous variables are z-scored prior to estimation. The binary state_level indicator is retained in its original 0/1 coding. Complete-case listwise deletion is used: only the 573 jurisdictions with non-missing values on all 10 indicators and 6 covariates are included.

Standard errors reported in the path tables (Table 6.2, Table 9.9) are conventional ML standard errors. Because many local jurisdictions within a state may share correlated unobservables, these conventional ML SEs understate uncertainty. Cluster-appropriate inference is implemented via a 1,000-iteration state-clustered bootstrap in Appendix C.6 (Table 9.17); the cluster-bootstrap SEs are 1.8–3.3 times the conventional ML SEs, and the 95% bootstrap CI for the headline capacity-timeliness path crosses zero. Closed-form cluster-robust SEs (sandwich estimator) are not provided by the semopy library; the bootstrap is the cluster-appropriate inference used throughout.

9.11 B.3 Full Parameter Tables

9.11.1 Measurement Loadings

Table 9.8 reports standardized factor loadings for the two-factor model.

Table 9.8: Two-factor model measurement loadings ($N = 573$)

Latent Construct	Indicator	Std. Loading	Std. Residual Var.	Note
Admin. Resources	Employment proxy	0.953	0.093	
Admin. Resources	Payroll proxy	0.992	0.016	
Admin. Burden Capacity	Rev. programs per staff	0.577	0.667	
Admin. Burden Capacity	Rev. disasters per staff	1.000	0.000	
Recovery Performance	Disbursed/obligated ratio	0.785	0.383	
Recovery Performance	Expended/disbursed ratio	-\$0.131	0.983	Weak negative loading
Recovery Performance	Fully expended share	0.988	0.024	
Recovery Performance	Program completion share	0.477	0.773	
Recovery Timeliness	Rev. duration of completion	0.545	0.703	
Recovery Timeliness	Rev. avg. program duration	1.000	0.000	

The expended-to-disbursed ratio loads negatively (\$-\$0.131) on Recovery Performance, indicating it does not cohere with the other three performance indicators. Sensitivity analysis dropping this indicator is reported in Appendix C.

9.11.2 Complete Structural Paths

Table 9.9 reports all structural regression coefficients for the two-factor model.

Table 9.9: Two-factor model complete structural paths ($N = 573$)

Outcome	Predictor	Unstd. Est.	Std. Beta	SE	p-value
Rec. Performance	Admin. Resources	-\$0.139	-\$0.170	0.037	< 0.001
Rec. Performance	Admin. Burden Cap.	0.104	0.077	0.060	0.086
Rec. Performance	state_level	-\$0.133	-\$0.038	0.200	0.505
Rec. Performance	Population	-\$0.015	-\$0.020	0.045	0.731
Rec. Performance	SVI Theme 1	0.008	0.010	0.053	0.879
Rec. Performance	SVI Theme 2	0.041	0.053	0.044	0.353
Rec. Performance	SVI Theme 3	-\$0.177	-\$0.226	0.041	< 0.001
Rec. Performance	SVI Theme 4	-\$0.022	-\$0.028	0.049	0.658
Rec. Timelines	Admin. Resources	0.046	0.080	0.025	0.072
Rec. Timelines	Admin. Burden Cap.	0.251	0.266	0.051	< 0.001
Rec. Timelines	state_level	-\$0.260	-\$0.106	0.133	0.050
Rec. Timelines	Population	0.009	0.016	0.029	0.760
Rec. Timelines	SVI Theme 1	-\$0.106	-\$0.194	0.036	0.004

Outcome	Predictor	Unstd. Est.	Std. Beta	SE	p-value
Rec. Timeliness	SVI Theme 2	-\$0.087	-\$0.159	0.030	0.004
Rec. Timeliness	SVI Theme 3	-\$0.014	-\$0.026	0.026	0.591
Rec. Timeliness	SVI Theme 4	0.006	0.011	0.031	0.846

9.11.3 Latent Covariances and Residual Variances

Table 9.10: Latent covariances and residual variances

Term	Estimate	Std. Est.	p-value
Admin. Resources variance	0.907	1.000	—
Admin. Burden Cap. variance	0.333	1.000	< 0.001
Rec. Performance residual var.	0.557	0.912	—
Rec. Timeliness residual var.	0.238	0.801	< 0.001
Rec. Performance ↔ Rec. Timeliness	-\$0.130	-\$0.306	< 0.001

R-squared values: Recovery Performance = 0.088, Recovery Timeliness = 0.199.

9.12 B.4 Measurement Discussion

Several measurement features warrant explicit discussion.

Two-indicator factors. AdminResources (employment, payroll) and RecoveryTimeliness (duration, average program duration) each have only two indicators. With two indicators per factor, reflective and formative specifications are empirically indistinguishable: the model is locally just-identified for the measurement component of each factor. The SEM is therefore best understood as a structured data-reduction and association-testing tool, not as a confirmatory latent-variable test.

Weak performance indicator. The expended-to-disbursed ratio loads at \$-\$0.131 on Recovery Performance, indicating it does not share the common variance captured by the other three

indicators. This may reflect accounting artifacts (expenditure timing relative to disbursement cycles) rather than substantive performance variation. A sensitivity check dropping this indicator is reported in [Table 9.11](#).

Reflective versus formative tension. The indicators may function as formative components (determinants of throughput capacity) rather than reflective manifestations (effects of a latent capacity trait). The reflective specification is adopted here because it enables conventional SEM estimation, but it is not independently validated. A rigorous validation would require confirmatory factor analysis on a hold-out sample with adequate power (conventionally $N > 200$ per group), comparison of reflective versus formative specifications using MIMIC or PLS-SEM, and convergent and discriminant validity testing. None of these steps were feasible given the current sample structure.

Parsimony-penalty tradeoff. The one-factor model has lower AIC (67.41) and BIC (219.69) than the two-factor model (74.36 and 239.69, respectively), despite worse incremental fit (CFI, TLI) reported in the main-text [Table 6.1](#). The two-factor model uses three additional parameters (the second factor loading, the factor covariance, and the second structural path set) that do not sufficiently reduce the log-likelihood to offset the information-criterion penalty. The two-factor model is retained as the reference specification because it captures the conceptually important distinction between resource levels and workload manageability, even though the information criteria do not favor it. This retention is theoretically motivated rather than statistically compelled.

9.13 B.5 Survival Analysis Specification

As a complementary analytical framework, a Cox proportional hazards model with time-varying covariates (Cox 1972; Collett 2015) is estimated on the same QPR data system but at the grantee-disaster level rather than the jurisdiction-profile level. The key specification differences are:

Unit of analysis. $N = 151$ grantee-disaster pairs, compared with 573 administering-jurisdiction profiles in the SEM. Smaller samples ($N = 142$ after government-type and size covariate drops; $N = 100$ for the staffing-indicator bridge) appear in specific sensitivity analyses and are documented at each table (see Appendix A.7).

Outcome. Time to 95% expenditure completion, measured in quarters. Programs that have not reached 95% expenditure are right-censored. Of the 151 grantee-disaster observations, 70 (46.4%) are completed events and 81 (53.6%) are right-censored.

Capacity covariates. Two financial flow ratios serve as time-varying capacity indicators: (1) the ratio of cumulative disbursed to cumulative obligated funds and (2) the ratio of cumulative expended to cumulative disbursed funds. Both ratios are clipped to $[0, 2]$ with a \$1,000 minimum denominator threshold. A one-quarter lag is applied to prevent reverse causality.

Static covariates. Government type, grant size (log), prior CDBG-DR experience, disaster year, population (log), disaster severity index, and baseline employment.

Standard errors. Bootstrap clustered standard errors are computed using 1,000 iterations with clustering at the grantee level to account for repeated observations within grantees.

Threshold sensitivity. The model is estimated across completion thresholds from 20% to 100% in 5% increments to assess whether results are threshold-dependent.

This specification differs from the SEM in three fundamental ways: it preserves within-grantee temporal variation rather than aggregating to a cross-sectional profile; it properly handles right-censoring through the survival framework rather than using observed duration as a latent indicator; and it uses financial flow ratios rather than staffing-scaled workload ratios as capacity measures.

Appendix C: Robustness and Sensitivity

9.14 C.1 Sensitivity Across Specifications

Table 9.11 reports the key structural beta (Admin. Burden Capacity → Recovery Timeliness) across fifteen specifications of the two-factor SEM, labeled by the Class taxonomy defined in main-text §4.1 (Class Ia measurement-preserving, Ic conditioning-set change, Ib sample-scope, II estimand-changing, II* estimand-changing with construct reassignment). Specifications use the 573-jurisdiction complete-case sample unless noted otherwise.

Table 9.11: Sensitivity dashboard across SEM specifications, labeled by Class (§4.1). Class Ia = measurement-preserving within same estimand; Class Ic = conditioning-set change (covariates alter the conditional estimand); Class Ib = sample-scope perturbation; Class II-C = capacity-operationalization change (denotes construct reassignment); Class II-O (outcome change) is reported separately in Appendix C.8. **Stability:** Stable = sign retained, magnitude $\geq 50\%$; Attenuated = sign retained, magnitude $< 50\%$ or not significant; Reversed = sign changes; Degenerate = estimation unstable. Descriptive annotations, not pass/fail.*

Specification	Class	N	CFI	RMSEA	β	Stability	Note
Measurement-model variants (Class Ia)							
Pooled two-factor (appendix reference)	—	573	0.915	0.081	0.266	—	Pooled reference for appendix comparability; main-text reference is local-only (N=543, $\beta=0.257$)

Specifi cation	Class	N	CFI	RMSEA	β	Stabilit y	Note
Drop weak indicat or	Ia	573	0.919	0.085	0.266	Stable	Remov es expend ed/disb ursed from Perfor mance
Emplo yment- only denomi nator	Ia	573	0.908	0.085	0.266	Stable	Workl oad = progra ms/em ployme nt
Residu alized burden	Ia	573	0.918	0.079	0.297	Stable	Burden indicat ors residua lized (Kron mal 1993)
Sampl e- scope subset s (Class Ib)							
Local jurisdic tions only	Ib	543	0.902	0.092	0.257	Stable	Reprod uces pooled result
State agenci es only	Ib	30	0.724	0.669	0.000	Degen erate	N too small
Nonzer o- QCEW subsam ple	Ib	111	0.918	0.079	0.132 (n.s.)	Attenu ated	Non- suppre ssed subsam ple; below

Specifi- cation	Class	N	CFI	RMSEA	β	Stabilit y	Note
Local- only AND nonzer o- QCEW Cohor t and maturi ty (Class Ib restric tions + Ic covari ate additio ns)	Ib	81	0.898	0.092	0.145 (n.s.)	Attenu ated	50% Decom poses state/lo cal confou nd
	Ib	463	0.899	0.089	0.236	Stable	Exclud e younge st portfoli os
	Ib	275	0.829	0.118	\$-\$0.244	Revers ed	Sign reversa l
	Ib	222	0.907	0.085	0.526	Stable	Strong er among experie nced grantee s
	Ic	573	0.915	0.080	0.105	Attenu ated	Below 50%; conditi oning- set
Maturit y-band control s							

Specifi cation	Class	N	CFI	RMSEA	β	Stabilit y	Note
Portfol io- scale control s	Ic	573	0.783	0.131	0.127	Attenu ated	change Below 50%; conditi oning- set change ; poor fit
Within - frame work capaci ty- operat ionaliz ation bridge s (Class II-C)							
Raw portfoli o counts	II-C	573	0.854	0.110	-\$0.443	Revers ed	Capaci ty-op change only
Financi al-ratio capacit y bridge	II-C*	573	0.926	0.073	-\$0.600 (p = 0.10)	Revers ed	Capaci ty-op change + constru ct reassign ment

Class Ia measurement-model variants are stable. Dropping the weak expended-to-disbursed indicator, using an employment-only denominator, and residualizing burden indicators on employment, payroll, and population all leave the point estimate within \$0.03 of the reference 0.266. The residualized specification survives the ratio-coupling test at 0.297, indicating the reference association is not purely a shared-denominator artifact in the sense of spurious ratio correlation (Kronmal 1993).

Class Ib sample-scope subsets reveal the QCEW proxy problem. The state-only model is degenerate at N = 30, and the local-only model (N = 543) reproduces the pooled β at 0.257.

Restricting to the 111 jurisdictions with nonzero QCEW employment attenuates the estimate to 0.132 and removes significance ($p = 0.16$). Restricting further to the 81 local jurisdictions with nonzero QCEW produces a similar attenuated estimate (0.145, $p = 0.22$), indicating the attenuation is not an artifact of the state/local compositional shift within the nonzero subsample. For the 80.6% zero-QCEW majority, the workload denominator reduces to an epsilon offset and the burden indicators function as raw counts rather than as genuine staffing-scaled ratios.

Cohort and maturity perturbations show that portfolio maturity partly explains the association. Excluding the youngest portfolios (Duration > 24 months, Class Ib) modestly attenuates the beta to 0.236. Restricting to mature portfolios only (>48 months, Class Ib) reverses the sign to $-\$0.244$; among mature portfolios, more manageable workloads are associated with *slower* observed timeliness, the opposite of the main-text finding. Conversely, restricting to jurisdictions with two or more disasters (Class Ib) strengthens the coefficient to 0.526, suggesting the association is stronger among experience-accumulated jurisdictions. Adding maturity-band or portfolio-scale controls (Class Ic conditioning-set additions) attenuates the pooled coefficient to 0.105 and 0.127 respectively.

Class II within-framework operationalization bridges isolate the role of operationalization by holding the SEM framework, unit of analysis, outcome measure, and covariates constant while changing only the capacity indicators. Replacing staffing-scaled workload ratios with raw log-transformed portfolio counts (Class II) reverses the sign to $-\$0.443$; this bridge is free of construct contamination because raw counts never appear as outcome indicators. Replacing them with financial-flow ratios (disbursed-to-obligated and expended-to-disbursed as reflective indicators of a capacity factor, Class II*) reverses the sign to $-\$0.600$; this bridge also involves construct reassignment because the same ratios serve as outcome indicators in the reference measurement model.

A conceptual caveat applies to the financial-ratio bridge specifically: the same disbursed-to-obligated and expended-to-disbursed ratios serve as Recovery Performance indicators in the reference SEM measurement model, but are re-cast as capacity indicators in this bridge. Two interpretations are possible. Under a strict construct-separation view, these ratios are outcomes only and their use as predictors in the bridge represents construct contamination rather than an alternative capacity operationalization. Under an operationalization-alternative view, financial-flow ratios can legitimately proxy administrative capacity (cf. the broader CDBG-DR literature that treats spending rates as throughput measures). The bridge result reported here should be interpreted under the second view while acknowledging that a portion of the sign reversal likely reflects the conceptual shift from outcome to predictor rather than a pure measurement difference. The raw-portfolio-counts bridge (row 14) is not subject to this contamination concern, because raw counts never appear as outcome indicators. The sign reversal on row 14 is therefore evidence that operationalization alone can flip the substantive sign.

The sign of the cross-sectional capacity-timeliness association in this data is meaningfully a property of how capacity is operationalized, not purely of the SEM framework.

9.15 C.2 Proxy Validation

Table 9.12 reports correlations between proxy variables to assess whether the QCEW staffing measures capture meaningful administrative variation or primarily reflect jurisdiction size.

Table 9.12: Proxy validation summary

Check	Sample	Estimate	Interpretation
Employment vs. payroll	Full (n=573)	0.945	High covariation as expected for co-scaled administrative inputs
Employment vs. population	Full (n=573)	0.169	Weak: staffing proxy does not collapse to jurisdiction size
Payroll vs. population	Full (n=573)	0.255	Moderate: some size-scaling in payroll
Employment vs. program count	Full (n=573)	0.170	Weak: staffing is not simply a function of portfolio volume
Employment vs. disaster count	Full (n=573)	0.036	Near zero: staffing is not proxying disaster exposure
External employment cross-check	States (n=27)	0.225	Weak: NAICS 925110 correlates modestly with total gov. employment
External payroll cross-check	States (n=27)	0.324	Moderate: payroll proxy is somewhat better validated externally
Residualized burden → timeliness	Coupling test	0.297	Positive: main finding survives arithmetic-coupling removal

The low correlations between QCEW employment and both population (0.169) and disaster count (0.036) are reassuring: the staffing proxy is not simply measuring jurisdiction size or disaster exposure. However, the modest external validation correlations (0.225 and 0.324) confirm that NAICS 925110 is an indirect proxy with substantial noise.

9.16 C.3 Portfolio Maturity Composition

Table 9.13 describes the composition of the 573-jurisdiction sample by portfolio maturity.

Table 9.13: Portfolio maturity composition ($N = 573$)

Maturity Band	N	State	Local	Median Programs	Median Disasters	Median Duration (months)
≤ 24 months	110	2	108	3	1	15.0
24–60 months	257	13	244	7	1	39.0
> 60 months	206	15	191	10	1	84.1

Young portfolios (≤ 24 months) are overwhelmingly local and have smaller programs. The attenuation of the burden-timeliness beta when maturity-band controls are added (Table 9.11) likely reflects this composition pattern: younger portfolios have had less time to demonstrate either throughput efficiency or completion, creating a mechanical correlation between maturity and observed timeliness.

9.17 C.4 Ratio Artifact Summary

Table 9.14 reports the frequency of ratio values exceeding 1.0 and a capped-ratio sensitivity check.

Table 9.14: Ratio artifact summary

Element	Value
Disbursed/obligated > 1.0	0 of 573
Expended/disbursed > 1.0	163 of 573 (28.4%)
Fully expended share > 1.0	52 of 573 (9.1%)
Program completion > 1.0	0 of 573
Capped-ratio SEM: CFI	0.886
Capped-ratio SEM: burden $\rightarrow$ timeliness β	0.266
Capped-ratio SEM: resources $\rightarrow$ performance β	-\$0.188

Expended-to-disbursed ratios above 1.0 occur in 28.4% of jurisdictions, likely reflecting timing mismatches between expenditure and disbursement reporting. Truncating all ratio indicators at 1.0 leaves the reference structural paths unchanged, indicating the reference results are not driven by ratio exceedances.

9.18 C.5 Cox Model Sensitivity and Staffing Bridge

9.18.1 Threshold sensitivity (single-spell specification with baseline covariates)

Table 9.15 reports a single-spell Cox proportional hazards model estimated across completion thresholds from 20% to 100%. Covariates are measured at baseline (the fourth reporting quarter, $q = 3$) to avoid outcome leakage; the event at each threshold is the first reporting quarter in which a pair's cumulative expended-to-obligated ratio reaches the threshold, with pairs that never reach the threshold right-censored at their final observed quarter. Duration is measured in quarters from the baseline quarter $q = 3$. Analytic sample is restricted to pairs that had not already crossed the threshold by $q = 3$ (N varies across thresholds for this reason).

*Table 9.15: Single-spell Cox model threshold sensitivity with baseline covariates ($q = 3$); event = first quarter cumulative expended/obligated reaches threshold. **Small-event caveat:** at the 90% threshold only 28 events inform the HR; at 95%, 20 events; at 100%, 7 events. Estimates at $\geq 90\%$ should be interpreted with caution given the sparse event counts and the attendant instability of the partial-likelihood estimator.*

Threshold	N	Events	HR(disb/ oblig)	p	HR(exp/ disb)	p	Concordance
20%	69	48	1.459	0.006	1.758	< 0.001	0.736
30%	79	53	1.611	< 0.001	1.432	0.002	0.758
40%	90	59	1.788	< 0.001	1.471	< 0.001	0.778
50%	95	60	1.954	< 0.001	1.364	0.001	0.772
60%	97	53	1.857	< 0.001	1.242	0.074	0.766
70%	100	46	2.002	< 0.001	1.310	0.034	0.748
80%	102	40	2.861	< 0.001	0.420	0.622	0.733
90%	106	28	1.933	< 0.001	1.329	0.007	0.805
95%	109	20	2.483	< 0.001	1.232	0.053	0.832
100%	110	7	2.574	0.002	0.006	0.355	0.913

The baseline disbursement-to-obligation ratio has a stable positive hazard-ratio relationship with subsequent completion across all thresholds: HR = 1.46 at the 20% threshold rising to HR = 2.58 at 100%, with $p < 0.01$ throughout. However, at higher thresholds ($\geq 90\%$) the event counts shrink rapidly (28 $\rightarrow$ 20 $\rightarrow$ 7), so the rising hazard ratios may partly reflect small-sample instability rather than a genuine threshold-dependent capacity effect. Jurisdictions with higher early-quarter disbursement ratios reach each cumulative-expenditure threshold faster than jurisdictions with lower early ratios. Concordance ranges from 0.74 to 0.91. This single-spell specification differs substantively from the time-varying Cox reported in the main text: the single-spell model asks whether *baseline* differences between jurisdictions predict subsequent time-to-completion, while the time-varying Cox asks whether *quarterly changes* in the lagged disbursement ratio affect the next-quarter hazard.

9.18.2 Reconciliation with the main-text Cox

The main-text Cox (Table 6.5) and the single-spell Cox (Table 9.15) answer different questions and reach different substantive conclusions on the same data:

- **Main-text time-varying Cox:** lagged (one-quarter) financial-flow ratios entered as time-varying covariates; $N = 151$ grantee-disaster pairs, 70 events and 81 right-censored at the 95% threshold (using `E` from `panel_time_varying.parquet`). $HR \approx 1.0$ on the lagged disbursement ratio ($p = 0.991$): quarter-to-quarter changes in the lagged ratio do not predict next-quarter hazard of crossing the threshold.
- **Single-spell baseline Cox (this appendix):** baseline financial-flow ratios (measured at $q = 3$) as static covariates; sample further restricted to pairs that had not yet crossed the threshold at baseline (hence varying N across thresholds). $HR \approx 2.0\text{--}2.9$ on baseline disbursement ratio ($p < 0.001$): jurisdictions with higher early disbursement ratios complete faster.

The two specifications are complementary rather than contradictory. The time-varying null means that the *dynamics* of the disbursement ratio (marginal changes from quarter to quarter) do not accelerate or decelerate the completion hazard. The single-spell positive effect means that the *level* of the disbursement ratio observed early in a jurisdiction's administrative trajectory does predict eventual completion. Both are properties of the data. The measurement-sensitivity reading of the paper includes this within-framework divergence in Cox specifications: the sign of the estimated capacity-outcome association depends on whether capacity is parameterized as a time-varying dynamic (null) or a baseline level (positive).

The corrected event coding also resolves a reconciliation issue with the event counts reported in this table vs. the main-text Cox. The main-text Cox draws the event indicator `E` directly from `panel_time_varying.parquet`, which encodes the first quarter each pair crosses the 95% threshold using cumulative expended-to-obligated ratio; this yields 70 events at the 95% threshold. The single-spell Cox in this appendix re-derives event timing from `qpr_quarterly.parquet` cumulative clean totals and restricts to pairs that had not yet crossed the threshold by baseline ($q = 3$); this yields 20 events at the 95% threshold from a restricted $N = 109$. Both event counts are correct under their respective definitions, and the event-count discrepancy reflects the different baseline restriction.

9.18.3 Staffing-bridge Cox

Table 9.16 reports a Cox proportional hazards model that holds the survival framework, censoring treatment, and grantee-disaster unit of analysis constant, substituting staffing-scaled workload indicators (z-scored programs-per-staff and disasters-per-staff) for the financial-flow ratios used in the main-text Cox model. The analytic sample is the 100 grantee-disaster pairs (48 events, 52 right-censored) for which staffing data can be merged on grantee name.

Table 9.16: Cox model with staffing/workload indicators as covariates ($N = 100$)

Covariate	HR	95% CI	p
z programs-per-staff	1.517	[0.56, 4.10]	0.412

Covariate	HR	95% CI	p
z disasters-per-staff	0.609	[0.16, 2.26]	0.459

Concordance = 0.39; partial AIC = 337.0. The Schoenfeld residual test does not reject the proportional-hazards assumption ($p = 0.87$ and $p = 0.93$). Neither staffing covariate is statistically significant. A concordance below 0.5 on the 100-pair subsample indicates that the staffing-scaled predictors order completion timing in the *reverse* direction of the survival-framework convention (higher predicted hazard with lower observed failure rate), which is consistent with the non-significant coefficients: the staffing indicators do not provide meaningful ranking information for completion timing here. The survival framework yields null capacity-outcome associations regardless of whether capacity is operationalized through financial ratios or staffing-scaled workload ratios: the SEM $\leftrightarrow$ survival divergence is not driven by capacity operationalization within the survival framework but by the joint consequences of the design differences between the two analytical frameworks (unit of analysis, outcome specification, censoring treatment, capacity operationalization, and covariate set).

9.19 C.6 State-Clustered Bootstrap Standard Errors for the SEM

Table 9.17 reports state-clustered bootstrap inference for the four main structural paths of the reference two-factor SEM. Bootstrap draws resample the 35 unique state-level clusters with replacement, preserving within-state dependence. Each of 1,000 bootstrap draws re-estimates the full two-factor SEM and records the standardized structural paths.

Table 9.17: State-clustered bootstrap inference for reference SEM structural paths (1,000 iterations, 35 clusters)

Path	Point β	Conventional ML SE	Cluster bootstrap SE	95% CI (cluster)
AdminReso urces $\rightarrow$ RecoveryPe rformance	\$- 0.170 0.037 0.080 -\$0.319, \$- \$0.004]			
AdminBurd enCapacity $\rightarrow$ RecoveryPe rformance	0.077	0.045	0.100	[\$-\$0.139, 0.243]
AdminReso urces $\rightarrow$ RecoveryTi meliness	0.080	0.044	0.084	[\$-\$0.083, 0.244]
AdminBurd enCapacity $\rightarrow$	0.266	0.054	0.177	[\$-\$0.129, 0.531]

Path	Point β	Conventional ML SE	Cluster bootstrap SE	95% CI (cluster)
RecoveryTimeliness				

Bootstrap-cluster standard errors are 1.8–3.3 times the conventional ML standard errors. The reference capacity-timeliness coefficient ($\beta = 0.266$, conventional ML $p < 0.001$) has a 95% bootstrap confidence interval of [-\$0.129, 0.531] that crosses zero. The AdminResources $\rightarrow$ RecoveryPerformance coefficient has a bootstrap 95% CI of [-\$0.319, \$0.004] that lies almost entirely below zero. Expanding from 200 to 1,000 iterations (R7 critique) leaves the substantive conclusion unchanged: the data do not distinguish the reference $\beta = 0.266$ from zero at conventional significance once within-state dependence is acknowledged. The script and per-replication outputs are at `data_work/diagnostics/sem_state_clustered_bootstrap_n1000.csv`.

A natural extension would report multiway clustering (e.g., state $\times$ disaster context). At the SEM unit of analysis (administering jurisdiction), each row aggregates across multiple disaster contexts in that jurisdiction’s portfolio, so disaster-level clustering is not naturally definable without collapsing portfolios onto a single dominant disaster. The complementary survival framework reports grantee-clustered bootstrap (Appendix B.5), which addresses within-grantee dependence at the appropriate unit of analysis. State-clustered bootstrap with semopy as implemented here is the cluster structure the reference SEM unit supports.

9.20 C.7 Observed-Variable Regressions

As observed-variable analogues to the latent-construct SEM, [Table 9.18](#) reports ordinary-least-squares regressions of each of the six outcome indicators on the four capacity indicators and the standard covariate set. All variables are z-scored; $N = 573$. Running the analysis as six separate regressions on observed outcome indicators avoids the construct-validity weaknesses of the latent two-factor SEM (two-indicator factors, boundary estimates, negative loadings, reflective/formative specification tension) while preserving the same substantive question.

*Table 9.18: Observed-composite OLS regressions of each outcome indicator on the four capacity indicators and the full covariate set (state_level, E_TOTPOP, SPL_THEME1–4). $N = 573$; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.*

Outcome indicator	$b(\text{Avg Employment})$	$b(\text{Avg Payroll})$	$b(\text{rev Programs/Staf f})$	$b(\text{rev Disasters/Staf f})$	R^2
Disbursed / Obligated	-\$0.277 ***	+\$0.073	-\$0.153 *	+\$0.221 *	0.115
Expended / Disbursed	-\$0.518 ***	+\$0.581 ***	-\$0.034	+\$0.026	0.052
Fully	-\$0.103	-\$0.066	-\$0.069	+\$0.114 *	0.102

Outcome indicator	$b(\text{Avg Employment})$	$b(\text{Avg Payroll})$	$b(\text{rev Programs/Staff})$	$b(\text{rev Disasters/Staff})$	R^2
Expenditure Share					
Program Completion Rate	-\$0.331 ***	+\$0.187	-\$0.067	+\$0.065	0.085
rev Duration of Completion (higher = faster)	+\$0.299 *	-\$0.183	+\$0.276 ***	-\$0.099 .	0.094
rev Average Program Duration (higher = faster)	+\$0.100	-\$0.009	+\$0.128 **	+\$0.187 ***	0.209

Three patterns in the observed-composite regression match the SEM structural patterns: (1) `rev_programs_per_staff` has a positive and significant association with both reverse-coded duration indicators ($b = +0.276$ and $+0.128$, both $p < 0.01$), mirroring the SEM `Burden → Timeliness` = $+0.266$; (2) `avg_employment` is negatively associated with three of the four performance indicators, consistent with the SEM `AdminResources → Performance` = $-\$0.170$; (3) the SVI theme coefficients (not shown here) associate with the same outcomes as in the SEM. That the observed-composite coefficient (0.276) is nearly identical to the latent-construct SEM coefficient (0.266) indicates the SEM is functioning primarily as dimension reduction rather than as latent-construct validation. The observed-variable regression and the SEM are complementary readings of the same associational structure: the regression trades latent-variable machinery for transparency at the cost of losing the covariance structure among outcomes, while the SEM trades transparency for the ability to summarize outcomes jointly under the acknowledged exploratory-decomposition assumption. Both read the same underlying signal; neither is a definitive specification on its own.

A narrower version of this analysis focused on financial-flow capacity indicators (disbursed/obligated and expended/disbursed) against Timeliness and Program Completion is consistent: $b(\text{disb/oblig, rev Duration}) = -\0.340 ($p < 0.001$); $b(\text{disb/oblig, Completion}) = +\0.365 ($p < 0.001$). *The pattern parallel to the within-SEM financial-ratio bridge* ($_{} = -0.600$ on Timeliness): higher disbursed-to-obligated ratios are associated with slower observed completion timelines, plausibly because jurisdictions with larger end-of-window disbursement shares tend to be mature portfolios that have had longer to execute.

9.21 C.8 Fixed-Horizon Outcomes Sensitivity

Table 9.19 reports cross-sectional regressions using fixed-horizon expenditure-share outcomes (cumulative expended-to-obligated ratio observed at quarters 8, 12, and 16 since first obligation). This outcome definition is cohort-invariant: it does not require a portfolio to reach 95% completion, so young and mature portfolios can both be observed on a common timeline.

Two versions of the fixed-horizon analysis are reported. The original (grantee-level) version aggregated quarterly expenditure data at the HUD Grantee level (N = 45 grantees) and merged with the SEM-level covariates, producing N = 32–36 after complete-case deletion. The **full-sample version** uses the 573-jurisdiction crosswalk (data_work/replication/jurisdiction_crosswalk.csv) to re-match F31 rows to SEM jurisdictions directly: state-agency jurisdictions pick up F31 rows whose Grantee equals the state code or state name; local jurisdictions pick up F31 rows whose Grantee State matches and whose normalized Activity Responsible Org contains the jurisdiction’s county name (with parish/borough variants). Quarterly flows are summed by jurisdiction-quarter and accumulated. The full-sample version yields N = 102–128 across the three horizons — substantially larger than the grantee-level version, though still less than the full 573 because upstream manual adjudication of city-level AROs is not recoverable for all jurisdictions (see Appendix A.5 Reproducibility Boundary).

9.21.1 Grantee-level fixed-horizon (N = 32–36, original)

Table 9.19: Grantee-level fixed-horizon expenditure-share outcomes regressed on capacity indicators and full covariate set. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Horizon	N	R ²	b(rev Programs/ Staff)	b(rev Disasters/S taff)	b(Avg Employme nt)	b(Avg Payroll)
q = 8	36	0.38	+\$0.148 *	-\$0.125 .	-\$0.121	+\$0.118
q = 12	36	0.32	+\$0.163 .	-\$0.121	-\$0.117	+\$0.110
q = 16	32	0.42	+\$0.126	-\$0.063	+\$0.010	-\$0.004

9.21.2 Reconstructed-panel fixed-horizon (N = 102–128)

Table 9.20: Reconstructed-panel fixed-horizon expenditure-share outcomes (N=102–128 jurisdictions, not the full N=573 SEM sample — see naming clarification below) regressed on capacity indicators and full covariate set using the 573-jurisdiction crosswalk and F31-reconstructed jurisdiction-quarter cumulatives (script: scripts/build_full_fixed_horizon_panel.py). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Horizon	N	R ²	b(rev Programs/ Staff)	b(rev Disasters/S taff)	b(Avg Employment)	b(Avg Payroll)
q = 8	128	0.077	+\$0.003	-\$0.062	+\$0.108	-\$0.010
q = 12	116	0.116	-\$0.001	-\$0.036	+\$0.066	-\$0.003
q = 16	102	0.167	+\$0.025	-\$0.076 .	+\$0.026	+\$0.040

The grantee-level and full-sample versions diverge substantively. At the grantee level (N = 32–36), *b*(rev Programs/Staff) is +\$0.148 at q=8 and significant ($p < 0.05$), consistent with the SEM Burden → Timeliness result. At the full-sample level (N = 102–128), the same coefficient is +\$0.003 at q=8, -\$0.001 at q=12, and +\$0.025 at q=16 — all within rounding distance of zero and none significant. The divergence is consistent with the interpretation that the pooled SEM’s positive capacity-timeliness coefficient is produced by the specific subset of grantees who have longer CDBG-DR administrative histories (and thus richer jurisdiction-quarter panels) combined with the ϵ -offset treatment of their QCEW zeros; the larger fixed-horizon sample dilutes that selection and the coefficient dissolves.

Naming clarification. The “reconstructed-panel” label distinguishes this analysis from the grantee-level (N=32–36) version reported above, but it is *not* the full N=573 SEM sample. The reconstruction is constrained by what F31 rows can be re-matched to SEM jurisdictions through the linker (state-level matching for state agencies; county-name matching for locals). 177 of the 573 SEM jurisdictions receive at least one F31 row through this re-matching; complete-case deletion on the fixed-horizon outcome plus the SEM covariate set yields N = 102–128 across the three horizons. A fully-reproducible fixed-horizon analysis at the N = 573 level would require recovering the upstream ARO-to-jurisdiction manual adjudication for the 1,708 unresolved labels, which is flagged in Appendix A.5 Reproducibility Boundary as a direction for future pipeline improvements. The reconstructed-panel fixed-horizon design addresses the maturity-confounding concern raised by the mature-portfolio sign reversal ([Table 9.11](#)) because the outcome is anchored at a fixed time since first obligation rather than at variable time-to-95%-completion.

9.22 C.9 SVI Vintage and Disaster-Year Sensitivity

9.22.1 State-level vintage sensitivity

CDC/ATSDR historical SVI vintages (2010, 2014, 2016, 2018, 2020, 2022) were downloaded from the ATSDR Feature Services and aggregated to state-level theme means. The reference two-factor SEM was re-estimated six times on the same 573-jurisdiction complete-case sample, substituting state-level SVI from each vintage while holding capacity indicators and outcome indicators constant.

Table 9.21: Reference SEM re-estimated with state-level SVI means from each CDC/ATSDR vintage (N = 573 reference sample)

SVI Vintage	N	CFI	RMSEA	Burden → Timeline ss	Resource s → Performance	Theme 1 (socioec.) → Timeline ss	Theme 3 (minority) → Performance
2010	573	0.901	0.094	+\$0.239	\$- 0.208 ** - \$0.215**	\$-\$0.256	
2014	573	0.904	0.096	+\$0.188	\$- 0.213 ** + \$0.460**	\$-\$0.226	
2016	573	0.911	0.095	+\$0.209	\$-\$0.216	+\$0.448	\$-\$0.213
2018	573	0.911	0.094	+\$0.221	\$-\$0.218	+\$0.449	\$-\$0.237
2020	573	0.908	0.097	+\$0.148	\$-\$0.227	+\$0.366	\$-\$0.204
2022	573	0.909	0.097	+\$0.170	\$-\$0.237	+\$0.167	\$-\$0.214

The capacity-timeliness coefficient is stable across SVI vintages (range \$+\$0.148 to \$+\$0.239). Resources → Performance and Theme 3 (minority/language) → Performance are also stable. SVI Theme 1 (socioeconomic vulnerability) → Timeliness changes sign between the 2010 and 2014+ vintages, reflecting a documented methodological revision in how CDC/ATSDR constructed Theme 1 between those releases. The substantive claim “socioeconomic vulnerability is associated with slower implementation” is therefore specific to the 2010 SVI vintage; later vintages reverse it.

9.22.2 Per-jurisdiction disaster-year SVI re-estimation

A more granular re-estimation assigns each jurisdiction the SVI vintage nearest-preceding its state’s earliest CDBG-DR disaster year (from the QPR panel) and matches at the county level (for the 543 local jurisdictions) or state level (for the 30 state agencies). Jurisdiction labels are parsed to extract state and county/parish/borough components; the state’s earliest CDBG-DR disaster year is computed from the QPR panel; the nearest-preceding CDC/ATSDR SVI vintage is then selected and county-level SPL_THEME1–4 values are looked up. The procedure resolves 572/573 jurisdictions to SVI values (Norfolk County, VA did not match because of Virginia’s independent-city nomenclature). Per-jurisdiction vintage assignments are: 2010 SVI for 476 jurisdictions (83%; states with earliest disasters pre-2014), 2014 for 11, 2016 for 57, 2018 for 25, and 2022 for 3.

Table 9.22: Reference SEM re-estimated with per-jurisdiction disaster-year SVI (N = 572, CFI = 0.903, RMSEA = 0.085)

Path	β	p	Comparison to midpoint-vintage reference
AdminBurdenCapacity → RecoveryTimeliness	\$+\$0.286	< 0.001	\$+\$0.266 reference; slightly stronger
AdminBurdenCapacity → RecoveryPerformance	\$+\$0.182	< 0.001	\$+\$0.077 (n.s.) reference; now significant
AdminResources → RecoveryPerformance	\$-\$0.162	< 0.001	\$-\$0.170 reference; stable
AdminResources → RecoveryTimeliness	\$+\$0.062	0.160	\$+\$0.080 reference; stable (n.s.)
SVI Theme 1 (socioec.) → RecoveryTimeliness	\$-\$0.331	< 0.001	\$-\$0.194 reference; stronger negative
SVI Theme 1 → RecoveryPerformance	\$+\$0.207	< 0.001	(n.s. in reference); new positive
SVI Theme 2 (household) → RecoveryTimeliness	\$+\$0.061	0.179	\$-\$0.159 reference; no longer significant
SVI Theme 2 → RecoveryPerformance	\$-\$0.317	< 0.001	(n.s. in reference); large new negative
SVI Theme 3 (minority) → RecoveryTimeliness	\$-\$0.095	0.057	new marginal
SVI Theme 3 → RecoveryPerformance	\$-\$0.116	0.019	\$-\$0.226 reference; attenuated
SVI Theme 4 (housing) → RecoveryTimeliness	\$+\$0.060	0.264	stable (n.s.)

Path	β	p	Comparison to midpoint-vintage reference
ess			
SVI Theme 4 → RecoveryPerformance	-\$0.162	0.003	(n.s. in reference); new negative

The capacity-timeliness coefficient is robust to the disaster-year SVI substitution ($\beta=0.286$ vs. 0.266 reference, $p < 0.001$ in both), and the AdminResources → Performance negative association is also robust ($\beta = -0.162$ vs. -0.170 reference). The SVI theme coefficients, however, reconfigure meaningfully. Under disaster-year SVI, Theme 1 (socioeconomic) is strongly negative on Timeliness and positive on Performance; Theme 2 (household composition) no longer predicts Timeliness but is strongly negative on Performance; Theme 3 (minority/language) on Performance attenuates. Together, the state-level vintage sensitivity and the per-jurisdiction disaster-year result indicate that SVI findings in cross-sectional CDBG-DR research should be reported with explicit vintage and granularity documentation. The substantive conclusions about which vulnerability dimensions matter for which recovery outcomes should be qualified as vintage-specific rather than treated as stable.

The disaster-year distribution of the 78 grantees with QPR first-disaster information is: earliest disasters 2001–2005 (7 grantees), 2006–2010 (14), 2011–2015 (26), 2016–2020 (27), and 2021–2025 (4). For the 57 grantees with earliest disasters in 2011 or later (73%), the disaster-year-appropriate SVI vintage is contemporaneous or post-disaster by at most 4 years. For the 21 grantees with earlier disasters, any SVI vintage post-dates the event by 6–15 years, so the SVI should be read as post-disaster recovery context rather than pre-disaster baseline.

9.23 C.10 QCEW Suppression Bounds Analysis

The bounds analysis operates on the 577-row recovered jurisdiction-profile file (the pre-complete-case frame, which differs from the 573-row complete-case SEM sample by 4 profiles dropped for missing SEM indicators — imputing QCEW zeros allows those 4 to be retained in the bounds SEM). Of the 577 profiles, 463 (80.2%) report zero QCEW employment in NAICS 925110; the corresponding share on the 573-row complete-case frame is $462/573 = 80.6\%$ (the body text reports the latter). Because these zeros are ambiguous between literal-zero and BLS-suppressed (Appendix A.3), the reference SEM’s ϵ -offset treatment of these zeros is a specific analytical choice rather than a neutral default. [Table 9.23](#) reports the reference two-factor SEM re-estimated with the 463 zero values imputed at a range of plausible alternative values, bracketing the BLS suppression-floor conventions: cell suppression typically applies when the industry-county-quarter cell has fewer than 3 establishments or fewer than about 10 employees. Translating those raw cell-count thresholds into the units of `avg_employment` (employees per thousand population) requires knowing the population over which the cell is reported. For the median zero-QCEW jurisdiction in the SEM sample (`E_TOTPOP` = 65 thousand), 3 employees correspond to `avg_employment` ≈ 0.046 employees per thousand population; 10 employees correspond to ≈ 0.15 . The imputation range 0.25–10 employees per thousand population therefore brackets the plausible suppression-cell magnitudes from a few employees per small county (low end) to dozens per medium county (mid range) to hundreds per large county (high

end). The payroll-per-employment ratio among non-suppressed observations is used to impute payroll proportionally.

A separate **ϵ -sensitivity scan** confirms that the choice of ϵ within the ϵ -offset family is analytically inconsequential. Refitting the reference two-factor SEM at $\epsilon \in \{10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}\}$ yields identical standardized structural-path coefficients to within numerical precision: Burden $\rightarrow$ Timeliness = +0.267, Resources $\rightarrow$ Performance = -0.180 across all four ϵ values (output: `data_work/diagnostics/sem_epsilon_sensitivity.csv`). The reason: z-standardization of the workload ratio removes the ϵ scaling, so what matters is the *relative ordering* of jurisdictions, which is determined by Num_Program differences among the suppressed-mass mode. The ϵ -offset treatment is the analytical commitment that matters; the specific ϵ value is not. The QCEW imputation bounds reported in [Table 9.23](#) therefore probe the consequential analytical choice (how to treat suppressed zeros), not the inconsequential one (which value of ϵ to use).

Table 9.23: QCEW suppression bounds sensitivity. Each row re-estimates the reference two-factor SEM with the 463 jurisdictions reporting zero QCEW employment in NAICS 925110 imputed at the specified value (employees per thousand population); payroll imputed proportionally using the non-suppressed median payroll-per-employment ratio. All other specification details (covariates, indicators, ML estimation) are held constant. Standardized structural path coefficients reported.

QCEW imputation (emp./K pop)	N imputed	Burden $\rightarrow$ Timeliness	Resources $\rightarrow$ Performance	Resources $\rightarrow$ Timeliness	Burden $\rightarrow$ Performance
0.00 (ϵ -offset reference)	463	\$+\$0.267	\$-\$0.180	\$+\$0.079	\$+\$0.169
0.25	463	\$+\$0.053	\$-\$0.122	\$+\$0.159	\$+\$0.079
0.50	463	\$+\$0.059	\$-\$0.110	\$+\$0.148	\$+\$0.075
1.00	463	\$+\$0.091	\$-\$0.082	\$+\$0.119	\$+\$0.066
2.50	463	\$+\$0.160	\$+\$0.041	\$+\$0.015	\$+\$0.035
(heuristic BLS- suppression- consistent floor)					
5.00	463	\$+\$0.052	\$+\$0.193	\$-\$0.152	\$+\$0.099
10.00	463	\$+\$0.024	\$+\$0.249	\$-\$0.223	\$+\$0.125

Two patterns are striking. First, the **capacity-timeliness coefficient (Burden $\rightarrow$ Timeliness)** ranges from +0.024 to +0.267 across the imputation scan, with the pooled ϵ -offset reference (+0.267) sitting at the top of the range. Under any plausible non-zero imputation of suppressed values, the coefficient is substantially smaller than the ϵ -offset reference (ranging +0.024 to +0.160). The reference $\beta=0.266$ is therefore not a centrally-located point estimate but a ceiling

value that depends on treating BLS-suppressed zeros as if they were literal zeros for the purposes of the workload-per-staff denominator. Second, the **AdminResources** → **RecoveryPerformance coefficient** flips sign: at imputations ≥ 2.5 (the heuristic BLS-suppression-consistent range — BLS does not publish a per-thousand-population suppression threshold; this floor is derived from the lowest non-suppressed observation for analytical comparison), the coefficient turns positive and reaches +0.249 at imputation = 10. This reverses the reference direction (-0.180) and undermines the characterization of the AdminResources path as “direction-stable under measurement-preserving perturbations” — it is direction-stable only *within the ε -offset specification family*. A Tobit-style censored-data SEM that treats the suppressed cells as left-censored at the BLS threshold would be the principled treatment; such an estimator is not supported by semopy and is documented as future work. The bounds analysis reported here is the feasible substitute and indicates that the two headline capacity paths behave differently: the burden → timeliness path remains positive across the full imputation range but attenuates sharply (from +0.267 to +0.024), while the resources → performance path reverses sign entirely (from -0.180 to +0.249). Direction stability is conditional on the ε -offset specification family for the resources path; magnitude stability is not achieved for either path.